Unsupervised Discovery of Non-Categorical L2 Error Patterns Using Wav2Vec2.0 Code Vectors†

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Abstract

L2 pronunciation is shaped by the interaction of two sound systems, which makes its identity more complex than a single phoneme category. This non-categorical nature calls for consideration of two additional factors in automatic speech assessment: 1) description units more granular than phonemes and 2) unsupervised methods to analyze data patterns free of external phonemic confinement. Past research on unsupervised error pattern discovery was motivated by practical ends to overcome annotation scarcity. They failed to factor in both aspects by either reverting to the categorical judgment or using the supervised and phoneme-prescribed feature phonemic posterior-gram (PPG). This work uses the self-supervised learning (SSL) feature, Wav2Vec2.0 code vectors, to address two requirements simultaneously while adopting the previous frameworks. Given the granular phonetic relevance of code vectors, we first verified their L2 discernibility by comparing featural usage between L1 (CMU ARCTIC) and L2 (L2 ARCTIC) speech. Frequency vectors mapping code vector occurrences were clustered to test their separability on account of nativeness. Subsequently, sub-categorical details of pronunciation errors were analyzed among segmentally identical samples in L2 Korean English NIA 037 data. After cataloging substitution segments detected by the model fine-tuned with L1 TIMIT, their corresponding code vector sequences were extracted by referencing the forced alignment result. We then derived dominant patterns of the sequences and compared them against L1 reference materials likewise constructed from TIMIT. Phoneme-code vector co-occurrence probability was used to infer phonetic attributes. Numerical records of code vectors and their clustering were used to infer pattern-wise Eucledian distance. The results showed that in addition to L2 discernibility, code vectors can quantify between-categorical error positions on the pronunciation spectrum. That is, variation patterns inferred from the phonetic and numerical standpoints concurred under the following linguistically interpretable common traits: (1) They formed a gradient error continuum along the changed articulatory value, (2) where the non-categorical intermediaries assumed opposite values in two codebooks. Moreover, (3) pattern distributions reflected the L1 influence and findings of existing L2 literature.

Keywords: *non-categorical, self-supervised learning (SSL), Wav2Vec2.0 code vector, granular*

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1. Introduction

The influence of L1 phonology in second-language speech learning is a well-narrated concept in L2 literature. Initial theories claimed unilateral influence, where perceptual space attuned to the native language sound system acts as a grid to simply filter a foreign acoustic input (Trubetskoy, 1939). The learnability, then, is solely determined by the degree of dissonance among structural oppositions in two languages (Lado, 1957). It was not long before scholars recognized a more bilateral nature of language learning, which extends to form a system of its own. Selinker's (1972) interlanguage hypothesis posits that linguistic organization in L2 results not only from L1 transfer but also from overgeneralized rules of the target language and experienced learning environment. In the speech domain, this hybrid dynamic corresponds to the adaptation of nonnative phonemes in the common L1-L2 phonological space.

Recent frameworks such as SLM (Flege, 1995) and PAM-L2 (Best, 1994) assume this mutual perceptual space where the existing L1 phoneme categories are adjusted, and the permissibility of new L2 categories is gauged. This complex interplay renders variability in phonetic knowledge that is more gradient than clear-cut. Given that production is guided by perceptual knowledge, learners' production value is also often between-categorical. Cantonese English speakers, for instance, may utter a variation of /t/ resembling both /t/ and /s/ (Li et al., 2018). Similarly, Korean English speakers are shown to pronounce /ð/, bearing characteristics of both /ð/ and /d/ (Kim, 2015). Such composite identities defy the earlier unilateral account as well as categorical evaluations using the canonical phoneme grid.

Nevertheless, existing technologies in mispronunciation detection and diagnosis (MDD) primarily rely on categorical judgment. Evaluation metrics utilize phoneme-level mismatch information: substitution, deletion, and insertion. For the substitution error, misrepresenting non-categorical traits (/n_l/) with a single phoneme (/n/ or /l/) will result in the problem of misdiagnosis (Mao et al., 2018). Still, the highly laborious nature of obtaining sub-segmental error labels holds research back from taking the much-needed granular approach. Annotating L2 speech, by itself, demands extensive language expertise to discreetly identify phonetic variations. The cost will increase further when the scope narrows to the subphonemic level, making manual performance impractical. The conflicting need between granularity and practical implementation has been settled by attempts to automate sub-segmental labeling using the framework of acoustic pattern discovery (Wang & Lee, 2013, 2015; Li et al., 2018; Li et al., 2020; Mao et al., 2018). The latter is a field in signal processing that aims to discover recurring signals for substituting manual labels. This tactic is employable to find patterns of L2 errors that are regular under systematic interaction. Pattern discovery has been applied to unsupervisedly identify mispronunciation even before realizing the need to account for its non-categorical nature. Lee & Glass (2012) and Lee et al. (2013) extracted alignment-based features between the native and nonnative utterances and constructed a binary classification problem for detection. Lee & Glass (2015) and Lee et al. (2016) selected phoneme confusion pairs by comparing acoustic distance among phone classes. They incorporated this information into the extended recognition network (ERN) for phonemic diagnosis.

As noted, early works on unsupervised error pattern discovery were mainly associated with resolving annotation scarcity. Addressing linguistic traits of non-categorical nuances was not its priority, as the discovered details were reduced to a binary or phonemic categorical interface. While the aforementioned five precursors address this concern to a certain extent, their underlying practical motive still confines them to a categorical framework. The analysis feature they worked with, phonemic posterior-gram (PPG), was intended to bring supervised phoneme-wise partitioning information to an unsupervised domain (Wang & Lee, 2015). PPG is a probability vector output from a model trained to classify phonemes. In a way, it is a literal rendition of the concept of the phoneme grid. Thus, while using

them to derive patterns suffices the goal of ostensibly instantiating variation, it is conceptually flawed. Defining a trait beyond phonemes with the same tool that limits the judgment is self-contradictory. In essence, previous literature relied on categorical knowledge for its practical emphasis on readily interpretable results.

In contrast, this work focuses on the conceptual grounding of deriving error patterns in an unsupervised manner. Unsupervised methods offer merit in addressing the compound identity of L2 pronunciation because they can escape the phonemic classification paradigm. Since they obtain knowledge solely based on data attributes, discovered patterns are free from external stipulation. Granted that the feature used to describe patterns is also more granular than and non-prescriptive of phonemes, we can observe phonetic variation from a new perspective. PPG satisfied the first criteria of being a granular unit yet fell short of the latter. The alternative feature we suggest here, the Wav2Vec2.0 code vector, meets both requirements. The feature falls under the broader category of self-supervised learning (SSL) representation, a conceptually fitting choice for the following reasons.

First, SSL representations are operative units of sound discovered through bottom-up inference. Self-supervised learning parallels human cognition, where general knowledge abstracted from data distribution is deduced to a specific applied context (Liu et al., 2022). During pre-training, the model learns representation encoding essential properties of unlabeled input, which lays the groundwork for understanding labeled contexts during fine-tuning. Thus, the feature exhibits representative information of language audio while being free from top-down phonemic stipulation. Moreover, this inference process aligns with the goal of acoustic pattern discovery: acquiring lexically meaningful units to classify and express speech content (Park & Glass, 2007). If the feature is already a rendition of discovered patterns, using them reinforces the research agenda.

Reflecting these conceptual grounds, linguistic relevancies of SSL representation have been surveyed in probing studies. They are known to capture various aspects of language, including phonological patterns, lexical meanings, and syntactic structures. Martin et al. (2023) demonstrated that HuBERT (Hsu et al., 2021) makes phonological distinctions between neutralized segments in aspiration. Pasad et al. (2024) investigated word-level properties in HuBERT (Hsu et al., 2021), Wav2Vec2.0 (Baevski et al., 2020), and WavLM (Chen et al., 2022) and found that layers containing lexical information differed according to the pre-training objective. Namely, the highest correlation was shown at a layer higher than the location of featural recovery. Shen et al. (2023) probed syntax encoding of the same SSL models used by Pasad et al. (2024). The work employed TreeDepth and TreeKernel probes and revealed that the encoding was the strongest in the middle to late layers, while it largely concerned lexical identity. Such relevancy further supports their use to examine language errors.

Among SSL representations, Wav2Vec2.0 code vectors were chosen for their confirmed phonetic relevance, granularity, and retrievability. In model documentation of Wav2Vec2.0 (Baevski et al., 2020) and its multilingual variant XLSR-53 (Conneau et al., 2020), linguistic properties of code vectors are examined. Baevski et al. (2020) plotted the conditional co-occurrence probability of phonemes with different code vector varieties and found that each feature specializes in representing separate phonemes. The correspondence relationship was also asymmetrical, with a single phoneme represented by multiple kinds. These findings suggest that while code vectors are phonetically relevant, they are finer-grained representations than phonemes, capturing distinct variational aspects within them. Meanwhile, Conneau et al. (2020) explored inter-language similarities of code vector representation in XLSR-53's pre-trained data. They constructed frequency vectors reflecting occurrences of codewords per language and clustered them in the mutual embedding space. Frequency vector plotting showed that the distance was smaller between close languages than between languages of disparate families. Such linguistic discernibility implies that code vectors are comprehensive features cognizant of shared acoustic characteristics. Lastly, Wav2Vec2.0 code vectors are retrievable thanks to the model-internal learning process.

Self-supervised learning of Wav2Vec2.0 consists of two main steps. Raw audio signal first enters a convolutional (CNN) feature encoder and becomes encoded into latent speech representation Z. Subsequently, the latent representation feeds into a transformer context network that outputs contextualized representations via masked contrastive learning. The learning objective is to mask a portion of the continuous Z and predict the correct (positive) representation from a set of (negative) distractors. Code vectors are latent representations quantized to form these positive and negative supervisory signals. As target labels guiding the entire supervision process, they compose the rudimentary acoustic units defined by the model. The use of supervisory signals for reconstructing masked features is a common architecture shared across different SSL models. While the quantized target is discovered within the model in Wav2Vec2.0, it can also be externally imposed. For instance, HuBERT (Hsu et al., 2021) uses externally clustered IDs as discrete baseline units. The large model uses features extracted from the base model's layers to generate cluster labels, which are difficult to directly access within its architecture. In this context. Wav2Vec2.0 offers a comparative advantage by providing direct access to its code vector index.

Ultimately, the suitability of Wav2Vec2.0 code vectors lies in their unsupervised, non-prescriptive nature and granular vantage point to access subphonemic variations. These characteristics preserve the conceptual integrity of unsupervised error pattern discovery in lieu of PPG. Hence, we aim to recreate previous works of non-categorical analysis (Wang and Lee, 2013, 2015; Li et al., 2018; Li et al., 2020; Mao et al., 2018) with our newly defined feature. Specifically, we will use Li et al.'s (2018) cluster sequence analysis (CSA) to extract representative frame-wise code vector index sequences among the same segment examples. This method, originally designed for PPG cluster IDs, will now be applied to code vectors to summarize sub-segmental nuances. Moreover, Kmeans clustering, a common similarity measure to derive sub-segmental identity in Wang & Lee (2013, 2015) and Li et al. (2018) and find non-categorical units in Mao et al. (2018), is used to infer the numerical attributes of the derived pattern.

In addition, linguistic probings of model documentation are employed to construct an interpretable interface. Unlike PPG that builds upon established phonetic knowledge, representations of code vectors are model-internal. Thus, the aforementioned probing methods are used as a medium to make linguistic conjectures. Language similarity probing in Conneau et al. (2020) is first applied to single-speaker data of L1 and L2 to test if the feature can make nativeness-based discernment. This frequency probing is a preparatory step to verify the code vectors' ability to describe subtle variations in L2 pronunciation. In the main error pattern discovery experiment, phonetic information probing in Baevski et al. (2020) is replicated using the same data to infer phonetic attributes of derived patterns.

With the combined usage of previous frameworks and code vector probings, this work limits the analytical scope to segmental substitution errors. The choice departs from Li et al. (2018) that applies analysis across all forced-aligned segments, necessitating an additional segmental error detection task. To this end, conventional MDD methods of calculating phoneme-wise alignment will be used. A pre-trained model used for code vector extraction was fine-tuned on phoneme labels to detect segmental errors, among which our analysis samples were selected. Code vector sequences corresponding to these samples were then extracted using forced alignment time stamps.

In the end, this experimental design intends to explore the range of variations code vectors can capture beyond the categorically imposed definition. It aims to answer the questions: 1) Does the feature encode L2 acoustics differently from L1? and if so, 2) How does it capture L2 pronunciation traits of non-categorical deviations? With the successful identification of its encoded linguistic properties, we may suggest the feature as a novel tool to conduct unsupervised MDD. Thus, we

focus on the inter-pattern relationship derived by code vectors and confirm if it concurs with phonetic understanding.

The rest of the chapters are organized as follows. Chapter 2 will overview three related research areas: acoustic pattern discovery, its previous application in MDD, and audio self-supervised learning, focusing on the Wav2Vec2.0 architecture we used. Chapter 3 will explain the proposed methodology, centering around two research questions raised earlier. Experimental details on the choice of data and analysis tools will also be expounded. Chapter 4 will lay out the results of the two experiments and summarize their main findings. Subsequently, Chapter 5 will discuss their implications, striving towards a comprehensive understanding across different analysis results. Finally, the summary in Chapter 6 will recalibrate the goal, findings, and contribution of this study.

2. Related Works

2.1. Acoustic Pattern Discovery

Earliest works in acoustic pattern discovery coincide with the task of unsupervised term discovery (UTD). Their objective was to automatically discover words and phrases from raw waveform to complement the zero resource setting. By comparing the utterance pair within the data, acoustic sequences of high similarity values are recorded as lexically meaningful units. This process comprises two steps outlined in the seminal work of Park & Glass (2007).

First, the pairs are aligned using the dynamic time warping (DTW) technique to calculate their warp path. The goal is to spot areas of lower distortion within alignment, which is synonymous with higher correspondence and similarity. A temporal constraint is additionally introduced to limit the global alignment to a local level, named hereafter as segmental DTW. Subsequently, local alignment pairs satisfying the distortion thresholds are united on a similarity basis through clustering. They are first constructed into an adjacency graph that encodes time and distortion information in nodes and edges. With all edges initially removed, nodes are iteratively merged in a bottom-up fashion to land on a pattern inventory.

While applying this pattern-matching algorithm, subsequent studies differ in the features used to denote sequences. If Park & Glass (2007) used mel-scale ceptstral coefficient (MFCC), later works opted for more advanced features with higher processing requirements. These range from acoustic embeddings in a fixed dimension (Kamper et al., 2016) to hidden markov models (HMMs) constructed per gaussian mixture model (GMM) cluster (Chan et al., 2011). Both alternatives initially take MFCC as an input, as the additional processing step aims at reducing signal variations. Among other options to normalize variations in MFCC, posterior-grams extracted from models trained on the feature were most prominent.

Posterior-gram is a probability vector over the designated model output categories. Namely, its kth dimension notates the associative probability of output with the kth category. Its normalizing property comes from a robust model training on diverse input data. In this context, Zhang & Glass (2010) applies Gaussian posterior-gram to the same pattern discovery pipeline in Park & Glass (2007) to resolve speaker dependency in MFCC. The GMM used for posterior-gram extraction was trained on multi-speaker corpus, rendering a speaker-independent feature. Aside from GMM, other instances of posterior-gram extraction models include dirichlet process GMM (DPGMM) (Ravi & Krothapalli, 2021), deep belief network (DBN) (Lee et al., 2013), acoustic phonemic model (APM) (Mao et al., 2018), MLP (Wang & Lee 2015), DNN (Li et al., 2018), and phonetic segment classifier (Li et al., 2020).

As a feature of classification outcome, the supervisory degree of posterior-gram varies depending on the model training scheme. If categories of probability assignment are externally given, it would be fully supervised. If they are acquired in an unsupervised way but bear some level of control, such as a pre-specified number of Gaussian components in Zhang et al. (2010), it would be semi-supervised. In contrast, Chan et al. (2011) automatically determine the number of GMM clusters using the minimal description length (MDL) principle, forming a fully unsupervised feature. Ravi & Krothapalli (2021) also obtained unsupervised posterior-grams by deriving classification components through a stick-breaking process in DPGMM.

2.2. Unsupervised Error Pattern Discovery

The working hypothesis of acoustic pattern discovery is that speech signals display enough regularity to statistically identify repeating patterns (Park & Glass, 2007). This idea extends to language learners' pronunciation errors that arise from the interaction of native and target sound systems. Given the parallel, acoustic pattern discovery has been applied to the task of MDD, drawing advantage of its unsupervised nature. In principle, the non-prescriptive discovery process exposes pronunciation traits uninhibited by categorizations.

2.2.1. Categorical Error Pattern Discovery

Nevertheless, with the final outcome orchestrated toward a categorical interface, the following four works lose this information. The research focus of this thread lies in either evaluating the presence of errors binarily (Lee & Glass, 2012; Lee et al., 2013) or phonemically decoding them (Lee & Glass, 2015; Lee et al., 2016). They also differ in the locus of comparison.

The former two make cross-speaker comparisons with native speech data to derive the peculiarity of non-native speech. Degree of misalignment between L1 teacher and L2 student utterances is calculated from relationship matrices into phone- and word-level features. These features, based on detailed acoustic measures, would have encompassed a gradient understanding of errors. However, the details are reduced to binary classification training of soft vector machines (SVM). Moreover, while the features used for utterance alignment were originally unsupervised in Lee & Glass (2012), the follow-up by Lee et al. (2013) opts for supervised DBN posterior-gram. This switch, despite satisfying the practical motive to improve error detection rates, further conflicts with preserving unsupervised granularities.

Meanwhile, the latter two studies inspect non-native data internally to locate patterns unique to individual learners. They begin by extracting sound segments corresponding to each canonical phoneme, referencing the forced alignment boundaries. These segments are initially compared grossly and then reinterpreted at a local triphone level. After selecting potential error candidates of the segment based on acoustic distance, the derived phoneme confusion pairs are subcategorized by their triphone identity. Nonetheless, ERN's greedy decoding process reduces the obtained rich sub-segmental nuances to a single segmental interpretation. More than identifying pattern traits reflecting acoustic distance, these works focused on retrieving the interpretable phoneme label to represent them.

2.2.2. Non-Categorical Error Pattern Discovery

The next five works partially share our interest in unsupervised methods. Procedurally, they aim for non-prescriptive discovery but are limited to employing a prescriptive feature phonemic posterior-gram (PPG). As noted, the supervisory nature of posterior-grams is determined by the level of external influence in defining classification categories. PPGs are fully supervised variants in the sense that categories are prescribed from human-annotated labels.

Nonetheless, the studies' goal of uncovering non-categorical characteristics of L2 pronunciation makes them direct precedents of the current research. There are two key aspects to this non-categorical approach. First, it self-discovers units of expression to elaborate on variation details rather than relying on a predefined phoneme set. Second, variations within the segment are preserved, unlike the greedy decoding that discards suboptimal trajectories. The following works rather collect these sound routes and analyze them to complete the gradience puzzle. While uncovering these paths in an unsupervised setting, they either limit the analytical scope to a single phoneme or expand on the whole utterances.

At the utterance level, Mao et al. (2018) and Li et al. (2020) used PPGs of the acoustic phonemic model (APM) and phonetic segment classifier to discover non-categorical units. Both works analyzed the shape of the posterior vectors to check how many phonemes a given signal is associated with. If more than one peak was observed among probability distributions, the associated sound was considered non-categorical, with its characteristics defined by peaking phonemes. They were notated with subjects of prominent posteriors, such as /eh_ey/ and /l_n/. Ultimately, non-categorical patterns were used to extend the single-category native phoneme set.

There were slight differences in how the discovery was carried out. Mao et al. (2018) used state-level PPGs averaged at the phoneme level, which were ultimately put through Kmeans clustering. Hereafter, the centroid of the clusters was the final analyzing component. Li et al. (2020) devised a segmental model to save the intermediary averaging step. Granted the refinement offered by the segment classifier, it also bypassed the clustering stage and analyzed individual output directly. Both PPG extraction models were trained on MFCC-phoneme label pairs, a practice shared by the other three works as well.

At a segment level, Wang & Lee (2013, 2015) and Li et al. (2018) examined variations within the given phoneme category. This approach addresses the state-level transitory details overlooked in the above macro analysis. To this end, either the same segment examples can be analyzed separately (Wang & Lee, 2013, 2015), or utterance-wise analysis can be applied at a segmental scope to specify variations among them (Li et al., 2018).

In the former case, frame-wise PPG sequences constituting each phoneme were restructured to summarize sub-segmental nuances. A hierarchical agglomerative clustering (HAC) algorithm was first applied to the featural sequence, where a pre-specified threshold was used to divide the resulting similarity tree into sub-clusters. Elements within each sub-tree were averaged and concatenated to form condensed feature representations. Ultimately, error patterns were derived by uniting representation variants of the same segment based on clustering similarity. Kmeans and GMM algorithm, guided by the MDL principle, were used for this final discovery.

In the latter case, frame-wise sequences subjected to error pattern discovery reflected a broader context, as they were notated with cluster IDs assigned to entire utterances. Frame-level PPG features of the dataset were clustered under the K-means algorithm, and the resulting cluster IDs were used to signify the identity of each phoneme-divided field of forced alignment. Since the ID sequence itself reflects contextual understanding, dominant patterns were extracted among the same-phoneme sequences without additional restructuring. Cluster sequence analysis (CSA) algorithm was used for this pattern derivation. The present study adopts the CSA algorithm for analyzing sub-segmental detail represented by code vectors. Hence, its finer details will be explained in the methodology chapter.

2.3. Audio Self-Supervised Learning

As noted, adopting the SSL representation Wav2Vec2.0 code vector is motivated by its unsupervised characteristic, which stems from the procedural resemblance of SSL to pattern discovery. In fact, both literatures cite the language development of infants as the inspiration behind each framework. Park & Glass (2007) notes that acoustic pattern discovery is a computational incarnation of how infants acquire lexical inventory from statistical properties of speech. Similarly, Liu et al. (2022) hint that audio SSL representations conceptually correspond to cognitive schemas governing the primary understanding of language acquired as an infant.

In the absence of explicit supervised labels, infants self-discover operating units to represent sound. Developed under this foundational assumption, audio SSL posits that representational units acquired via self-supervision are generalizable enough to correspond with prescribed labels. That is, knowledge learned performing the upstream tasks may be deduced to match the specificity of the downstream context. Thus, before utilizing the former's universal property, it would be worthwhile to survey the underpinnings of the upstream tasks that make up this trait.

There are two guiding principles governing the extrapolation of general knowledge. Paired views of 1) the same object must correlate, while 2) the views of different objects must be disassociated (Liu et al., 2022). The first principle alone forms the generative modeling goal of prediction. By maximizing the correlation between similar positive pairs, a condensed latent representation is created. The informativeness of this feature is further enhanced through a reconstruction task, where learned global contexts are exploited to estimate local targets by leveraging their mutual relationship. If the locus of reconstruction is sequential from the input, the objective is autoregressive predictive coding (APC), while masked predictive coding (MPC) predicts the output of the masked regions. Meanwhile, the model may learn distinctive representations by distinguishing the pairs of the same objects from the different ones. This discriminatory task entertains two principles simultaneously by minimizing the distance between the former and maximizing the distance between the latter. The combinatory learning objective is reflected in the alignment and uniformity term of the contrastive loss function (Liu et al., 2022).

In this context, Wav2Vec2.0 (Baevski et al., 2020) assumes a composite modeling goal that makes contrast-based predictions over masked regions. Namely, it operates on contrastive MPC. As briefly explained, its architecture comprises two-staged encoder networks along with the quantization module to form supervisory signals. The initial CNN encoder outputs the condensed latent information, which is then fed to the transformer encoder to make local guesses from global inference. Along with the contextual reference, it uses quantized latent of the prediction time indices to learn the distinction between the true value and false distractors. The learning function reflects the desired relationship between these two references as it maximizes the relative similarity of the masked context vector to the true quantized latent compared to its similarity to a set of distractors.

As the learning proceeds to iteratively readjust the similarities, two types of representations encode the final training result of the upstream task: quantized latent code vectors and context vectors. We chose code vectors as the analysis feature because they align better with our conceptual motivation. First, they are less supervisory in nature compared to context vectors formed under the weak supervision of self-created targets. As the constituents of these supervisory targets, code vectors are essentially pseudo labels serving as a proxy of human annotation. This makes them more rudimentary units. Second, in relation to training parallel with cognitive faculty, code vectors bear closer resemblances to an abstract schema. Context vectors are also involved in fine-tuning, which conversely implies their closer connection to the deductive logic aspect of cognition. Third, in terms of the dual modeling goal that is both generative and discriminative, features can be interpreted as either being more comprehensive or goal-specific. As context vectors engage in fine-tuning, they are concerned with substantializing sounds with prescribed labels. In contrast, code vectors are comprehensive representations agnostic of prescribed utilities at the interface.

There were also practical reasons behind this choice. Probing surveys show that while context vectors handle higher-level meaning, code vectors are more versed in capturing lower-level traits. Martin et al. (2023) demonstrated that the CNN encoder processing latent representations can better encode phonological information than transformers. Accordingly, the complexity of transformers may overpower lower-level distinctions. This understanding aligns with Pasad et al. (2024)'s finding that the higher word-level information is better represented at layers beyond the local unit the model operates on. Above all, the relevancy of lower-level feature code vectors to phonetic units has been explicitly tested. Fig.2 and 9 cite graphs from model documentation of Baevski et al. (2020) and Conneau et al. (2020). The Wav2Vec2.0 architecture is capable of learning dis-

crete representations that are distinctive (yet more granular) across phoneme categories (Fig.2) while generalizable towards mutual phonetic traits across languages (Fig.11). Code vectors' discreteness also makes them more tractable than continuous context vectors. This is crucial for our unsupervised approach, as it eliminates the need for an intermediary processing step to tract patterns. Such a step would introduce prescribed engagement, which we aim to avoid.

3. Methodology

Fig.1 outlines the overall framework of this research. To answer the first question of whether code vectors possess L2 discernability, featural inventories of individual L1 and L2 speakers will be surveyed. After confirming this baseline eligibility, they will be used to identify gradient characteristics of pronunciation errors beyond their segmental identities. Since our research focuses on substitution errors that defy categorical diagnosis, segmental error detection precedes the main error pattern analysis. Note that this is an optional step to limit the analytical scope. In practice, the unsupervised sub-segmental discovery is applicable to any segment of interest.

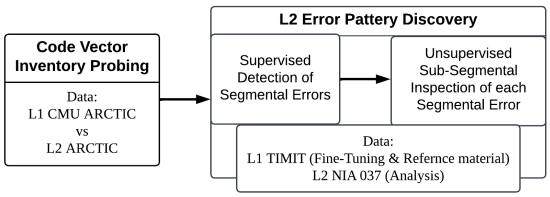
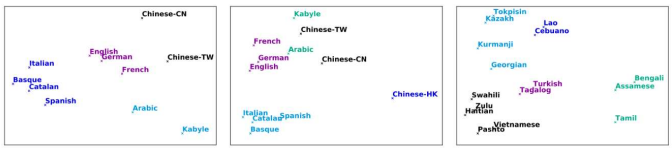


Figure 1. Overall Framework

3.1. Code Vector Inventory Probing

To validate code vectors as an analysis feature, we must verify their ability to distinguish between native and non-native speech. This task can be achieved by comparing the featural set used to encode L1 and L2 speaker data. Code vectors are formed through product quantization that divides the high-dimensional latent into multiple subvectors and maps each onto the closest codeword from respective codebooks (Jegou et al., 2010). The chosen codewords are then concatenated to form the final format of discrete representation. In Wav2Vec2.0, quantization uses 2 codebooks, each containing 320 codeword entries. Since it concatenates two entries from respective codebook, featural usage can be viewed from the perspective of 1)How frequently each entry is used and 2)How frequently each concatenated form appears.



(a) CV-12 model. (b) CV-12 model + zh-HK. (c) BABEL-17 model.

Figure 2. Language Similarity Probing in Conneau et al. (2020)

The former approach was taken in the language similarity probing of XLSR-53 (Conneau et al., 2020), where cross-lingually trained codewords shared distribution in line with language similarity. Fig.2 shows visual plotting of the frequency vector that accounts for codebook entries per language data. As can be noted, a cluster coordinated with the same color resembles a family of similar languages, such as Italian-Catalan-Spanish and Swahili-Zulu-Haitan. By replicating the same methodology used here, we aim to test if code vectors' inter-language discernibility extends to intra-language nativeness-based discernment. The target of frequency calculation is now set as individual L1 and L2 speakers as the analytical scope narrows to within a single language. Fig.3 visually demonstrates how frequency vectors were constructed at the 1) entry-level.

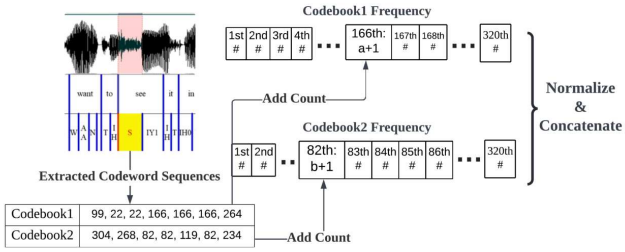
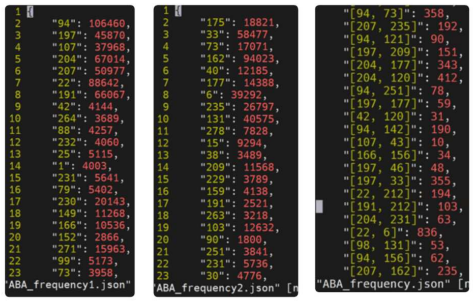


Figure 3. Construction of Codeword Frequency Vectors. a and b are the hypothetical number of occurrences before adding a count

The construction process first involved forced-aligning speaker files and selecting the most representative indices of the phoneme-divided field. In the above example, the 166th index and 82nd index are each selected from the first and the second codebook. The aggregated frequencies per entry were recorded as a key-value pair in a JSON dictionary. The first two figures in Fig.4 illustrate the frequency recording at the entry-level. The raw count was then normalized and transformed into a probability vector of size 320*1 per codebook. As not all 320 entries were in use, empty slots were filled with a zero value. Ultimately, the two normalized vectors were concatenated to represent the codeword usage of each speaker. As can be noted, the maximal number of employable codewords is fixed to 640 (2*320) kinds. This was not the case for the 2) pair-level analysis.



Codebook 1 Codebook 2 Codeword Pair

Figure 4. Frequency Recording in JSON Dictionary - L2 ARCTIC speaker ABA

Since two codewords pair up to represent sound frames, the maximal number of possible representations squares the number of codebook entries, i.e., 320 * 320 = 102.4k (Baevski et al., 2020). Within these possibilities, the set of codeword pairs employed by individual speakers varied. This resulted in a mapping problem as not all speakers shared the same counting bins to map occurrences. We resolved the inconsistency by creating a frequency vector on the union of all available paired indices in the entirety of the dataset. Namely, frequency recording was first made per individual, which was later adjusted by adding non-present key indices paired with 0 value. For the quantizer of the pre-trained model we used, the number of concatenated vectors outputted from each speaker ranged from 892 to 3133 kinds. These amounted to 5712 varieties across the L1 and L2 datasets. Hence, frequency vectors of 892 to 3133 dimensions were adapted to 5712-dimensional space, whose raw counts were likewise normalized into a probability. The rest of the construction process follows the prior method. The JSON frequency recording of the same speaker surveyed earlier is presented in the rightmost figure of Fig.4. We further illustrated the adjustment process in Fig.5. The sound

segment for index recording is identical to the example in Fig.2. Unlike the entry-level example, however, counting bins are tailored to the set used by each speaker. In the given case, it consists of 1080 pairs. Consequently, the 4632 elements not present in the union set are added with the value of 0.

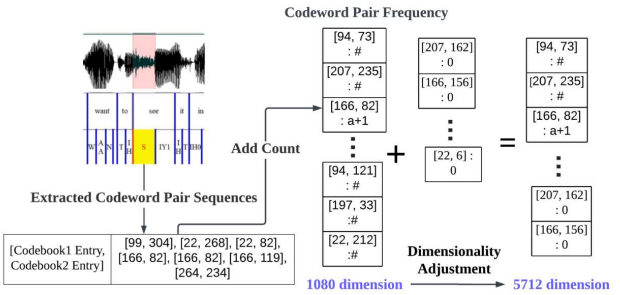


Figure 5. Construction of Codeword Pair Frequency Vectors. a is the hypothetical number of occurrences before adding a count

Meanwhile, the diverse range of featural usage prompts us to make detailed cross-speaker inspections. Pair-level occurrences deserve local analysis in addition to mass comparison in vector space, as they directly reflect usage patterns. This rationale was further enhanced after observing the idiosyncrasies that exhibited analytical potential beyond binary grouping. To uncover the specificity of each speaker's data, a set of paired indices they used was first recorded. This set was analyzed in terms of 1) how much they overlap with other L1 or L2 speakers and 2) their relative size to other speaker's inventory. When observing mutuality, both raw overlapping numbers and ratios were considered. The ratio was calculated by dividing the size of the mutual code vector inventory by the total amount employed by each speaker. Fig.6 exemplifies how individual code vector inventory is incorporated into deriving the mutual count and ratio metrics. If the demonstration concerns two L1 and L2 speakers, actual implementation concerns a total of 24 speakers, forming a 24*24 matrix.

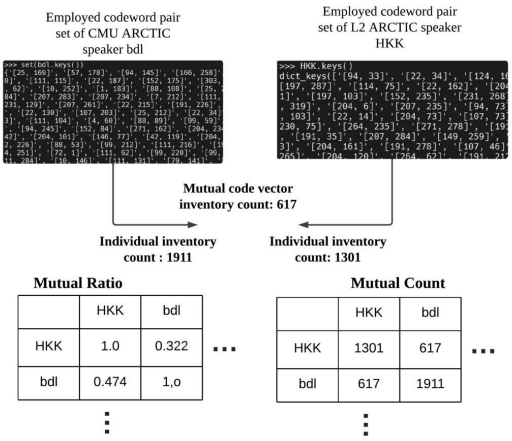


Figure 6. Formation of Comparative Matrices for Mutuality Analysis

While Fig.6 concerns pair-level analysis only, the clustering of frequency vectors was a discernibility measurement applied at both levels. We followed the outline in Fig.2 by first performing Kmeans clustering and visualizing the result after projecting the data onto two-dimensional space using principal component analysis (PCA). The cluster number K was 2, as the expected result was to have L1 and L2 speaker data develop into separate clusters. This separation would reflect the code vector usage differences.

3.2. L2 Error Pattern Discovery

As mentioned, the present study accompanies an auxiliary supervised detection task to focus our unsupervised endeavor on segmentally defined substitution. We confined our analysis to substitution errors since they present a complex categorical diagnostic problem. Deletion and insertion errors, in contrast, do not possess corresponding segments to substantialize non-categorical characteristics. To catalog a list of substitutions submitted for pattern analysis, we followed standard practice in MDD of comparing recognized phonemes with ground truth labels. The former is taken as how the learner uttered speech, while the latter refers to the canonical pronunciation.

To generate phonemic transcription, the same pre-trained model used for code vector extraction was fine-tuned with a downstream task in automatic speech recognition. After running recognition on the target L2 analysis data and aligning it with the expected label sequence, areas of mismatch were recorded and sorted from the highest to lowest occurrences. In this list of catalogs, substitution samples that satisfied the criteria of frequency and phonetic relevancy were selected. The frequency was considered to garner enough samples to discover patterns representative of the L1-L2 pair. Moreover, we had to filter linguistically irrelevant error types that likely resulted from the recognizer defect. To this end, we referred to existing L2 literature on Korean English (Yang, 2013; Kim & Rhee, 2019; Choi & Oh, 2021) and consulted previous work on ASR-based error analysis (Hong et al., 2021) to raise the objectivity of our analytical choice. Fig.7 gives examples of recorded instances and selected error types annotated with relevant theoretical groundings cited from the literature.

Error Type	Ground Truth	Recognized Result	Frequency	Theoretical Grounding
Substitution	*Z*	*S*	*11204*	*Limited Fricative Inventory in L1*
Deletion	T	***	7668	
Substituion	*L*	*R*	5549	*Posited as allophones in L1*
⋮				
Insertion	***	IH	4791	
Deletion	N	***	3386	

Figure 7. Examples of Detected Segmental Errors. *selected substitution errors are italicized.*

To analyze the selected samples, their constitutive code vector sequences were retrieved as indices. We first logged the time stamp of the erroneous segment by comparing the forced alignment result with the recognition-ground truth alignment. This comparison was viable by employing the same phoneme set for two procedures. Fig.8 demonstrates how the temporal information of falsely recognized phonemes was documented. At the upper left is a mismatch instance between IH and IY. The lower left shows forced alignment TextGrid, from which a list of time stamps for code vector extraction is gathered.

Figure 8. IH to IY Substitution Sound Retrieval

Subsequently, code vector indices of the corresponding time frames in each audio file were extracted as in Fig.9. The recordings were

at a pair-level, with codeword indices of two codebooks notated as [codebook1 index, codebook2 index]. We subject this sequence set to the pattern analysis method of Li et al. (2018). Among four non-categorical pattern analyses of previous frameworks (Wang & Lee, 2013, 2015; Li et al., 2018; Li et al., 2020; Mao et al., 2018), we chose this strategy for the following reasons.

First, as we aim to limit the focus to designated errors, methods specialized towards a single segment type (Wang & Lee, 2013, 2015; Li et al., 2018) were more suitable than those aimed at cross-segment characteristics (Mao et al., 2018; Li et al., 2020). Contemplating between the former two, we realized that code vectors encode acoustic information hovering over the entire training data. Thus, the holistic consideration of Li et al. (2018) seemed to better align with the nature of our used feature. Lastly, Wang & Lee (2013, 2015)'s approach to deriving condensed representation involved pre-specified cutoff thresholds, which varied per phoneme type. Such external involvement conflicted with our unsupervised interest.

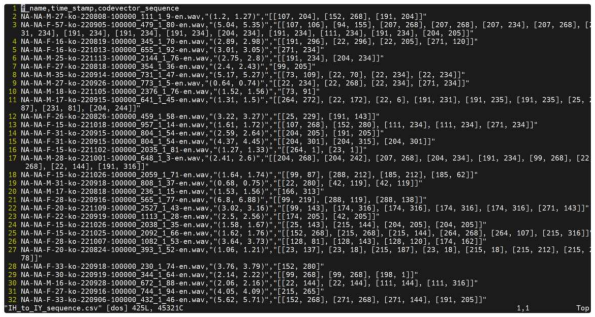


Figure 9. Code Vector Sequences of IH to IY substituion

The precise details of CSA consist of three steps. The first step involves removing minor presences among sequence constituents below the designated threshold. Accordingly, we filtered the index that showed single occurrences between disparate varieties. The second step furthers the refinement by summing adjacent re-occurrences to one, leaving a single cluster ID per transition. The third column of Fig.10 demonstrates the outcome. In the present study, we reflected this scheme by reducing consecutive duplicates into one. Lastly, dominant sequence types are selected based on frequency. We heuristically decided on the type selection threshold as 3. In most analyzed cases, calculating frequency on a refined set resulted in the three most dominant patterns, each represented by a single paired index.

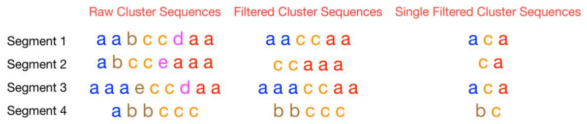


Figure 10. Cluster Sequence Analysis (CSA) strategy in Li et al. (2018)

These three code vector indices were taken as internally derived patterns and were interpreted from two perspectives, one from numerically learned values of training and the other from shared values of linguistic understanding. Since the model was pre-trained on L1 data, we constructed the reference materials for interpretation using an L1 corpus. The data used here is identical to the one used for fine-tuning, keeping the consistency of analysis across segmental and sub-segmental scope. For each purpose, we recorded 2) what phonemes each codeword (pair) statistically represents and 1) the raw numerical values of code vectors as opposed to indices. The former is a revisit to the initial probing experiment performed in Wav2Vec2.0 (Baevski et al. 2020). In the paper, the conditional probability of phoneme distribution per discrete latent is plotted by counting its co-occurrence with human-annotated phoneme boundaries. The same method of counting co-occurrences is applied, with our L1 reference/finetuning data also being identical to the corpus used here.

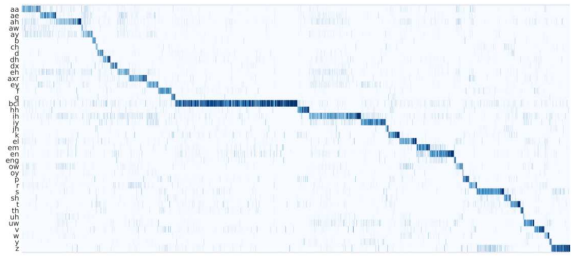


Figure 11. Phonemic Probing in Baveski et al. (2020)

At each time frame, the encoded paired index is documented alongside the corresponding phonemes. This resulted in a dictionary mapping phonemic occurrences per index, as shown in Fig.12. The integer values of the dictionary are transformed into probability, recreating a cross-section of Fig.11 along the vertical axis. The phoneme set used for probing, however, is slightly different. If Fig.12 collapsed the original notation to 39 phonemes, we utilized all 61 phoneme-like-unit (PLU) available in the dataset, including the begin and the end marker #h. The choice was to enable more granular judgment in line with our research goal. We demonstrate the relationship between our reference material, Fig.11, and 12 in Fig.13.

One thing to note is that the reference materials were created at both the (codeword) entry and pair level. In Fig.12, recordings of co-occurrences are executed on two separate codebook entries, as well as the concatenated form of these indices. The reason for surveying entry-level information is to account for cases where L2 pattern indices use unprecedented pairing in L1. In the former inventory probing, we spotted that index pairs used in L2 often do not coincide with the used set in L1. The only way to decode the pattern attributes in these scenarios is to take references from individual indices.

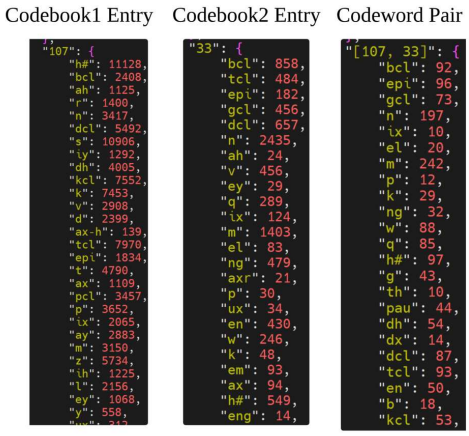


Figure 12. Frequency of Co-occurring Phonemes per Code Vector

The numerical recording follows this guideline as well. In addition to the concatenated pairs, we recorded separate codeword entries per codebook. This amounts to three vector spaces in Fig.14, each representing the first and the second codebook and their joint space. The paired entries were additionally Kmeans clustered into 39 groups, which corresponds to the size of the phoneme set used for model fine-tuning and forced alignment. Namely, it reflects the categorization

criteria for segmental error detection, a fitting choice to probe if two pattern indices form separable identities. As each entry has 384 dimensions, the original dimensionality of the vector spaces is 384 for individual codebooks and 768 for codeword pairs. We performed PCA and reduced these high-dimensional spaces to 2 dimensions for visualization in Fig.14. In the joint space, clustering preceded dimensionality reduction to preserve data integrity.

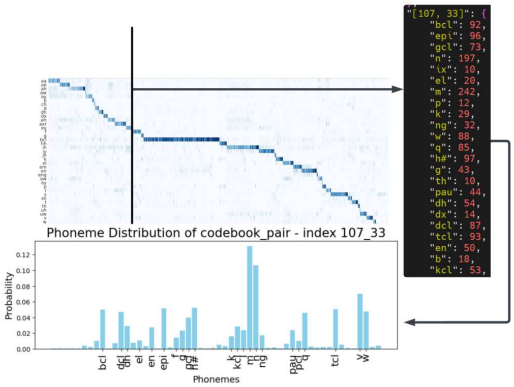


Figure 13. Relationship between Linguistic Reference Material and conditional plotting in Baevski et al. (2020)

The number of codeword entries used in the L1 data we created reference material from was 67 (codebook1) and 215 (codebook2). These varieties combined to form 2022 kinds of concatenated pairs. The size difference implies that it is the pairing of codewords that creates a differentiable feature, thus deserving detailed inspection in 3.1. Fig.16 summarizes the entire L2 error pattern discovery process.

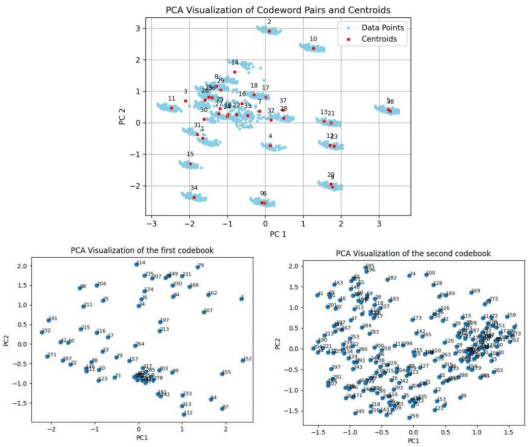


Figure 14. Code Vector Spaces in L1 Reference Data

3.3. Experiment

3.3.1. Data

As briefly noted in Fig.1, two different L1-L2 datasets were used for the experiments 3.1 and 3.2. Each task came under different data selection criteria. Fig. 15 summarizes the choice we made.

For inventory comparison, we needed read-speech of L1 and L2 using the same prompt script to regulate content-wise variables from interfering with the code vector frequency. Garnering speaker-level recording was also crucial as we aimed for local analysis within a single language, unlike the global language-wise comparison in Fig.2. To observe trends differentiating a particular demographic in cross-speaker inspection, we needed a sufficient amount of data per individual. Accordingly, the chosen corpora are speaker-wise recordings over the same 1132 list of the Arctic reading prompt.

For L1, the CMU ARCTIC is a phonetically balanced, single-speaker English database primarily designed for speech synthesis research (Kominek et al., 2003). Among the available 7 speakers (bdl, slt, jmk, awb, rms, clb, ksap), 5 speakers of North American accent were used (bdl, rms, jmk, rms, clb). Speakers of other accents, namely Indian and Scottish English, were excluded as this research primarily concerns US English. For the L2 counterpart, the L2-ARCTIC (Zhao et al., 2018) corpus encompasses non-native English speech from speakers of 6 L1 backgrounds: Arabic, Mandarin, Korean, Vietnamese, Hindi, and Spanish. Our used version, 5.0, includes a Praat TextGrid format of phonemic transcription generated via Montreal Forced Aligner (MFA). As noted, code vector occurrences are recorded by the most representative index in the field of the respective phoneme. Thus, this forced alignment information served as the baseline for frequency calculation. Since CMU ARCTIC did not provide a pre-generated TextGrid, we replicated the L2 ARCTIC generation process using the identical tools in MFA.

Experiment	3.1	3.2
L1	CMU ARCTIC (5 speakers: bdl, slt, jmk, rms, clb)	TIMIT (fine-tuning: train split, reference material: train & test split)
L2	L2 ARTIC v5.0 (all 24 speakers of 6 L1 backgrounds)	NIA 037 Korean English Speech data for Educational Purposes (sentence - level utterances in train split)

Figure 15. Overview of the Used Datasets

Meanwhile, to arrive at generalizable findings in error pattern analysis, it was more beneficial for the data to include a diverse range of speakers than being specialized to a single individual. The NIA037 Korean English speech data for educational purposes (Han et al., 2024) well satisfied this diversity requirement. The corpus included pronunciations of L2 learners across different proficiency levels, ages, and gender. Consequently, we chose sentence-level utterances from the train split as our main analysis data. Phrase-level utterances were discarded due to difficulties in automation. MFA requires manual annotation of each speech chunk to perform phrase-level alignment, which conflicted with our unsupervised pipeline.

Multi-speaker acoustics was also a desirable standard for L1 corpus used for both fine-tuning and constructing reference material. For the former, a diverse training set could create robust speech recognition. For the latter, referencing the traits encompassing diverse speakers enhances the objectivity of interpretation. Additionally, a reliable timestamp to document the corresponding phoneme accurately in Fig.12 was preferred. This led us to choose the TIMIT (Garofolo et al.,1993) corpus, containing transcription at the human-confirmed phoneme range. In terms of diversity, it comprises speech from 630 speakers across eight North American English dialects.

3.3.2. Implementation Detail

Framework: All SSL-related works and the model fine-tuning were performed under fairseq[1]framework, a sequence modeling toolkit developed by the Facebook AI Research (FAIR) group.

Pretrained Model: For both code vector extraction and model fine-tuning, an identical variant of the Wav2Vec2.0 model is used. The version used here is the LARGE architecture trained on LibriVox (LV-60k) data. This is also the variant whereby the original Wav2Vec2.0 discrete latent analysis was performed. The choice more or less stemmed from the confirmed linguistic relevance here.

Fine-tuning: Using the TIMIT train split, the chosen pre-trained model was fine-tuned under the parameter setting of CTC criterion,

[1]https://github.com/facebookresearch/fairseq

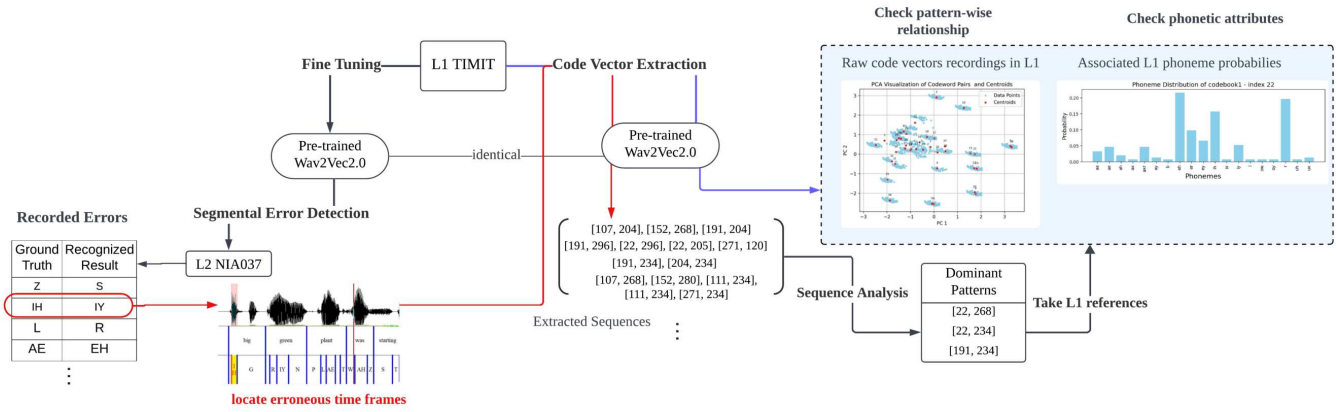


Figure 16. Overview of L2 error pattern discovery

40000 max update, 3e-4 learning rate, update frequency=4, adam optimizer (betas: 0.9Õ.98, eps:1e-08), tri stage learning rate scheduler (ratio 0.1,0.4,0.5), mask probability 0.65, mask channel probability 0.5, mask channel length 64. The final validation phoneme error rate for the fine-tuned model was 1.63%.

Grapheme to Phoneme Conversion: The phoneme set used for both fine-tuning labels and forced alignment were the 39 ARPAbet symbols. For fine-tuning, the set was derived by excluding the sentence stress numeric from the g2p tool kit [2]. For forced alignment, excluding the stress marker in English (US) ARPA dictionary v3.0.0 of Montreal Forced Aligner resulted in the same union of symbols. Keeping the representation format identical for model training data and forced alignment was crucial to consistently spot the area of mispronunciation designated by the automatic recognition result. These 39 symbols include: AA, AE, AH, AO, AW, AY, B, CH, D, DH, EH, ER, EY, F, G, HH, IH, IY, JH, K, L, M, N, NG, OW, OY, P, R, S, SH, T, TH, UH, UW, V, W, Y, Z, ZH.

Forced Alignment: Forced alignment is used in both 3.1 and 3.2. In 3.1, we have briefly mentioned that L2-ARCTIC v5.0 comes with pre-generated alignment that we will replicate to the comparative L1 corpus CMU ARCTIC. The tools employed here were the English (US) ARPA v.3.0.0 dictionary and acoustic models in MFA. To maintain the consistency of measurement, we used the same method for segmenting boundaries for code vector extraction in 3.2.

Clustering and Visualization tools: Clustering of paired code vectors and visualization of individual codewords were performed with the Facebook AI Similarity Search (FAISS[3]) library. This library facilitated efficient clustering of high-dimensional data on the GPU. In 3.2, the Matplotlib library in Python was used to visualize the conditional phoneme distribution.

3.3.3. Evaluation Metric

In frequency vector clustering in 3.1, we used the Davies-Bouldin Index (DBI) to additionally evaluate the cluster quality. DBI compares the relative sizes of inter-cluster separation and intra-cluster cohesion. It is calculated as the ratio of the average distance between cluster centers (numerator) and of data points within a cluster (denominator). Hence, a lower value marks superior performance. In Equation (1), N, c, and S(igma) each represent the number of clusters, cluster centroid, and the average standard deviation within each cluster.

$$DBI = \frac{1}{N}\sum_{i=1}^{N}\max_{j\neq i}\frac{\sigma_i+\sigma_j}{d(c_i,c_j)} \quad (1)$$

[2]https://github.com/Kyubyong/g2p
[3]https://github.com/facebookresearch/faiss

4. Results

4.1. L1 to L2 Code Vector Usage Comparison

We begin with analyzing the mass comparison of frequency vectors. Fig.17 shows the plotting of codeword frequency vectors across 29 speakers of L1 (5) and L2 (24) corpora. Note that clustering was performed prior to the principle component analysis (PCA). Although the dimensionality is reduced for visualization, these vectors are originally 640 dimensions. Each dimension represents the frequency of codewords constituting two codebooks of size 320. Speakers of the frequency vectors are annotated next to the plotted location, while each data point is color-coordinated according to its cluster membership. As can be noticed from the color scheme, the cluster grouping coincides with the division line between native and non-native speech.

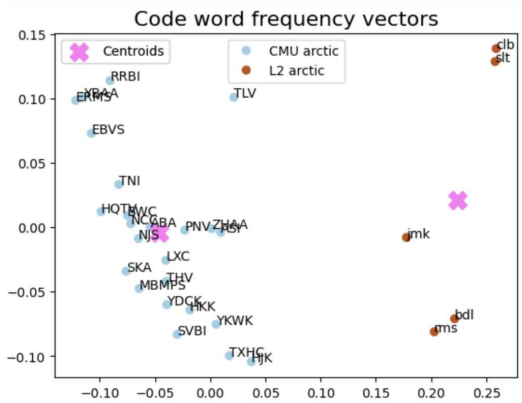


Figure 17. Visualization of Codeword Frequency Vectors Clustering

Nevertheless, the separation is not the most marked, especially when compared to the inter-language frequency plotting in Fig.2. This is expected since the variation based on nativeness is a lot more subtle than the acoustic distance between different language groups. To parse the subtle differences, we should take a step beyond the entry-level probing and examine the frequency of the combined format. Accordingly, Fig.18 displays the plotting of the pair-level examination. The pair-level vectors are, as mentioned, of 5712 dimensions. Each dimension accounts for the frequency of an element constituting the union of used varieties across 29 speakers. This dimensionality is likewise reduced to two for the sake of visualization. The clustering of codeword pair frequency vectors, indeed, displays a clearer separation between the L1 and L2 speaker groups. The DBI score improved from 1.038 (entry-level clustering in Fig.17) to 0.82 (pair-level clustering in Fig.18). This improved metric implies that the acoustic traits of

L2 learners are more distinctly encoded in the combined format, prompting us to conduct detailed cross-speaker comparisons in a paired setting.

Fig.19 corresponds to the result of the speaker-to-speaker mutuality analysis outlined in Fig.6. We begin the comparison with ratio to observe overall trends. The heatmap intersection shows the proportion of code vectors used by both speakers relative to each speaker (listed in the top row)'s unique code vector inventory. This ratio is calculated by dividing the number of shared vectors by the total number of unique items employed by the speaker listed horizontally. The first 24 speakers in the row are from the L2 ARCTIC corpus, while the last 5 are from CMU ARCTIC.

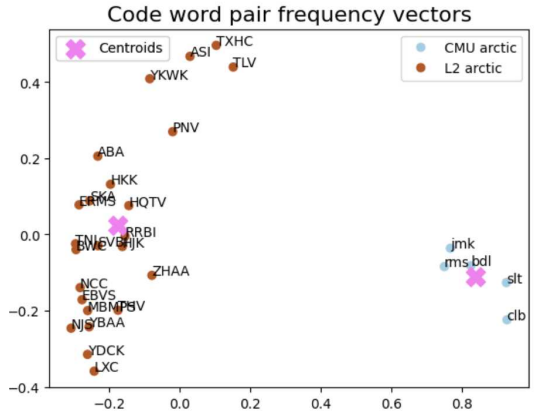


Figure 18. Visualization of Codeword Pair Frequency Vectors Clustering

The most conspicuous trend is that the sharing rate between native and non-native speakers is far below the rate within each speaker group. Intersections of the L1 and L2 speakers display a blue hue, indicating lower similarity than the intersections internal to each group (that, in contrast, exhibits a red hue). This reflects the former clustering result in that the used inventory overlaps more strongly among each clustered group.

The second notable point is that among the heat map intersections of L1 and L2 speakers, the shared ratio calculated based on the CMU ARCTIC speakers (right) is lower than the ones calculated based on the L2 ARCTIC speakers (bottom). Every speaker pair generates two intersection points on the comparative matrix, with different speakers serving as the reference points. When the inventory size of L1 speakers is used as the denominator for calculating ratios, the resulting values were lower than when the mutual size was divided against the numbers concerning L2 speakers. This suggests that L1 speakers generally employ a more diverse range of features compared to L2 speakers. We further examined this matter by directly referring to the raw mutual inventory size in Fig.20.

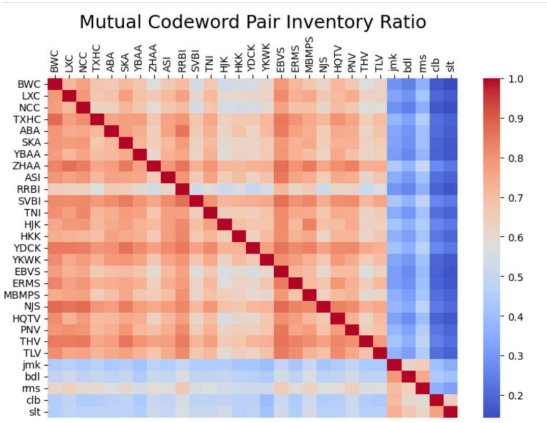


Figure 19. Mutual Inventory Ratio per Speaker Pair

The heat map below reflects the raw count of the mutual inventory shared between two speakers at the intersection. Since the values are not normalized, the mutuality ratings are not as ostensibly comparable as in Fig.19. Nevertheless, one noticeable trend is that within each speaker group, the upper left L2-ARTIC grids have darker shades associated with fewer tokens compared to the bottom right CMU ARTIC grids. This sharing rate asymmetry aligns with our hypothesis of a higher number of codeword pairs utilized in native speech. Meanwhile, the self-referential diagonal grid of L2 speakers assumes a diverse range of numerical values. The question followed whether there is an underlying factor determining how many pairs would be used to encode each non-native speaker's L2 speech.

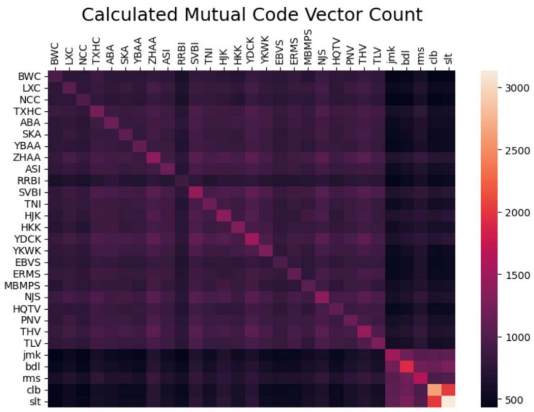


Figure 20. Mutual Inventory Count per Speaker Pair

To observe trends overlying L2 speakers, we first gathered their demographic information from the corpus documentation (Zhao et al., 2018). Fig.21 shows the language proficiency of speakers who participated in corpus construction upon its first release. Based on the TOEFL iBT scores, we divided these speakers into three groups representing varying degrees of speaking proficiency. The lower-level group consists of speakers SKA, EBVS, BWC, and LXC, whose scores range below 90. The intermediate-level group, consisting of speakers YBAA, RRBI, and TNI, had scores between 90 and 109. Finally, the higher-level group included speakers with scores equal to or greater than 110 (HKK, YDCK, NJS). Expanding on the former ratio analysis, we surveyed the specific number of code vectors these 10 speakers shared with 5 native speakers in CMU ARCTIC.

Speaker	L1	Gender	TOEFL iBT
HKK	Korean	M	114
YDCK	Korean	F	110
BWC	Mandarin	M	80
LXC	Mandarin	F	86
YBAA	Arabic	M	100
SKA	Arabic	F	79
EBVS	Spanish	M	70
NJS	Spanish	F	110
RRBI	Hindi	M	91
TNI	Hindi	F	99

Figure 21. Demographic Information of L2 ARTIC v1.0 speakers in Zhao et al. (2018)

Fig.22 charts the precise overlapping numbers. Though subject to minor variations, a general trend follows that the shared tokens increase in proportion to articulation adeptness. The chart records the overlapping amount in the increasing order of proficiency from top to bottom. Accordingly, a progressive increase in the number is observed from the low-level group (blue) to the intermediate-level group (green) and, finally, to the high-level group (red). Such minute detail is a finding not easily spotted in the ratio analysis (Fig.19), emphasizing the importance of considering both ratio and count

for comprehensive understanding. The variation in overlapping degrees further translates into the overall amount of used paired tokens. Fig.23 shows the inventory size of the 15 speakers pertinent to proficiency analysis with their known proficiency levels and nativeness annotated in the leftmost column. Note that the remaining 14 L2 ARCTIC speakers, who were not part of the initial release, were excluded from the analysis as their meta-information was unavailable.

	L2 / L1	jmk	bdl	rms	clb	slt
Low	BWC	440	476	597	447	446
	LXC	471	504	613	497	517
	EBVS	449	499	591	491	501
	SKA	491	542	657	513	520
Intermediate	YBAA	501	555	683	521	531
	RRBI	479	520	634	489	505
	TNI	529	572	671	569	598
High	HKK	550	617	685	604	607
	YDCK	627	696	771	702	728
	NJS	584	660	789	625	645

Figure 22. The Shared Amount of L2 Speakers against L1 Speakers

Referring to the relative size of the speaker inventory brings the overall analysis results into perspective. With the exception of speaker RRBI, the L2 speaker's proficiency rating correlates with the diversity of the used features. This trend is further supported by comparing L2 speakers to native speakers. Given that CMU ARCTIC speakers, representing native-like proficiency, utilize the most diverse pairs, we can establish a linear relationship between proficiency level and code vector diversity. This relationship ranges from low-proficiency L2 speakers (TOEFL < 90) to intermediate-proficiency L2 speakers (TOEFL $91 - 109$) to high-proficiency L2 speakers (TOEFL > 109) and, ultimately, to native speakers.

One possible reason behind this phenomenon is that to fully utilize representations encoding English, one has to be phonetically aware of the sounds in the language. This awareness is marked by the utilization ratio of the code vector inventory acquired in the L1 standard. If the amount used by the native speakers is the full range of available acoustics, the less proficient one is at articulating constitutive sound units, the less amount of inventory in use there will be. The language here does not particularly pertain to English but rather applies to any variant used during pre-training, whereby the quantization weight is learned.

L1	jmk	bdl	rms	clb	slt
	1516	1911	1633	2637	3133
L2 - high	HKK	YDCK	NJS		
	1301	1491	1387		
L2- Inter	YBAA	RRBI	TNI		
	1123	893	1180		
L2 - low	BWC	LXC	EBVS	SKA	
	973	1074	964	1080	

Figure 23. Total Count of Code Vector Inventory per Speaker

4.2. L2 Error Pattern Discovery Result

4.2.1. Detected Segmental Errors

We previously explained in 3.2 that a list of substitution errors will be selected prior to submitting them for non-categorical analysis. If Fig.7 gives an exemplary demonstration, this sub-section will chronicle the full-fledged selection process. Out of error samples displaying high frequency, a total of 13 error types concerning 6 substitution trajectories were selected. Previous studies on Korean learners' English pronunciations were referred to to this end. Fig.24 summarizes the selection result. As can be noted, the types of phonetic sounds targeted in our substitution samples encompass three categories: fricatives, liquids, and vowels. Accordingly, we will narrate the linguistically relevant reason behind sample selections by each.

Phonetic Category	Substituion Trajectory	Error Type	Phonetic Category	Substituion Trajectory	Error Type
Fricative	voiced > voiceless	**Z** to **S**	Vowel	Tenseness + / -	**IH** to **IY** **IY** to **IH** **AE** to **EH** **EH** to **AE**
	continuous > non-conitnuous	**DH** to **D** **V** to **B** **F** to **P**		Mid > Low	**AH** to **AA**
Liquid	Laterality + / -	**L** to **R** **R** to **L**		Diphthong > Monophthong	**EY** to **EH** **OW** to **AO**

Figure 24. Summary of Selected Substitutions for Pattern Discovery

Fricatives constituted the highest occurring error types out of all learners' errors. The selected four substitutions (Z to S, DH to D, V to B, F to P) are also reported to be the most noticeable consonantal variations produced by Korean Learners in the ASR-based corpus analysis of Hong et al. (2014). Korean fricative inventories are relatively limited in size compared to the rich inventories in English. The differing granularity of phonemic contrast increases the distance between the target phonemes and the closest approximates in the learners' L1. This disparity likewise increases the difficulties in production. Voiced fricatives are particularly challenging, with the added dimension of voicing distinction absent in the Korean consonantal system (Smith & Swan, 2001). Concerning two aspects of difficulties, we documented examples of two substitution routes.

The first case, Z to S, changes the voicing identity from voiced to voiceless while maintaining the place and manner aspects of the articulation. We may infer the cause from the fact that Korean alveolar fricatives are all voiceless (/s/ and /s=/), reducing the articulatory sensitivity to voicing distinctions in this particular sound class.

The latter cases commonly involve a shift towards homorganic plosives. In DH to D, V to B, and F to P, the manner of articulation changes from fricative to plosive while maintaining the place and voicing aspect of articulation. These errors are likely fostered by a comparatively richer plosive inventory than fricative in Korean. Consequently, the acoustic distance between the English plosive sound and the existing variants in the speaker's mother tongue is closer than the distance rendered from fricatives. This may cause learners to approximate more distant fricative sounds to acoustically familiar plosive counterparts. Such tendency has been observed in perceptual experience in Schmidt (1996), where DH/ð/, D/d/, and F/f/ were each mapped to Korean lax plosives /p/, /t/, /k/.

In summary, our selected examples of fricatives demonstrate how the learners' difficulties in adjusting to disparate sound systems result in morphing foreign acoustics in familiarized ways. Nevertheless, as we will observe, the L1 influence does not land on a homogeneous outcome as prescribed in previous literature.

Liquid substitution between the rhotic R and lateral L is likewise attributed to the difference in L1 and L2 sound systems. The bidirectional changes can be viewed under the same lighting. Unlike English, exhibiting two distinct lateral and rhotic liquids phonemically, Korean liquid inventory consists of only one element. Rhotic (tap) and lateral [l] are allophonic variations of the single phoneme /l/ (Kim & Ree, 2019). This inconsistency leads to confusion in producing English liquids since the two modes of articulation do not always show complementary distribution as in the learners' L1.

Moreover, the rhotic of English /ɹ/ differs from the Korean allophone flap, employing an articulatory measure that is unfamiliar

to Korean learners. In comparison, /l/ is an unmarked sound that occurs in both sound systems. Koo (2012) notes that such contrast in systematic complexity creates greater difficulty in learning the pronunciation of the English rhotic compared to the liquid.

Meanwhile, the core finding of Kim & Rhee (2019) was that the articulation accuracy of English liquids improves as a function of learners' proficiency. The representativeness of liquid pronunciation as an evaluative indicator supports our choice. Since liquid substitution errors provide a window to assess overall speaking proficiency, they become a relevant target for sub-segmental analysis.

Vowel errors selected for the analysis concern three different linguistic factors. The first is the absence of a tense-lax distinction in the Korean vowel system. In IH to IY, IY to IH, AE to EH, and EH to AE, the learner confuses tense and lax vowels with their opposite counterparts. Given the bidirectional relationship between the two phonemes, substitution directions have been studied in pairs (Koo & Oh, 2001; Yang, 2010, 2013). Among these paired segments, Yang (2010) notes that lax IH and tense AE, which are not present in Korean, pose greater articulatory challenges than IY and EH. The perceptual experiment in Tsukada et al. (2005) reflects this skewed tendency. Their results show that while IH and AE were often confused with IY and EH, the reverse did not hold—namely, IY and EH were mostly correctly identified. The asymmetrical relationship between AE and EH is further highlighted in the production analysis of Koo & Oh (2001). Accordingly, Korean English pronunciations of AE exhibit format values closer to the mid-vowel range of EH than to its canonical low vowel range. Such tendencies are worth noting, as they are later reflected in our pattern analysis.

The second substitution trajectory directly occurs in vowel height. In addition to the previously mentioned IH-IY and EH-AE, Tsukada et al. (2005) suggest that AH-AA is another prominent English vowel pair that native Korean speakers find difficult to distinguish. This finding is consistent with Koo & Oh's (2001) observation that AH and AA are pronounced in similar mid-back areas of the vowel space. Consequently, we may anticipate that within vowel substitution patterns, articulation will tend toward AH, which canonically represents mid-vowels. When mapping foreign AH to familiar native phonemes, AH bears between-categorical traits encompassing the boundaries of Korean /a/ and /ʌ/. As Korean /a/ forms a close acoustic distance from English AA, it was detected as the most dominant substitution segment with 4452 occurrences. The AH to AA substitution was also identified as a prominent vocalic variation in other speech recognition-based error detection studies (Hong et al., 2014).

The final category of vowel errors concerns the production characteristics of diphthongs. Diphthongs involve a swift transition between different articulatory positions of two vowel sounds within a single syllable. This poses a unique challenge for language learners, as their production requires rapid articulation across movement trajectories. Studies show that the less proficient a speaker is, the greater the distance of tongue movement, which leads to the segmented production of the two components (Park, 2001; Oh, 2013). For the English diphthongs EY/eɪ/ and OW/oʊ/, failure to articulate both segments within the time limit results in substitution with adjacent monophthongs. Cho & Jeong (2013) analyzed cases where Korean learners reduce OW to AO. In a similar vein, Choi & Oh (2021) discovered that Korean English speakers tend to produce EY as EH, which also reflects the closer acoustic distance of EH to the Korean /e/.

4.2.2. Discovered Sub-Segmental Patterns

In 3.2, we outlined a two-step pattern analysis process: L2 internal pattern derivation followed by external interpretation using L1 reference materials constructed from two perspectives. If the initial internal inspection discovers the three most dominant pattern indices, the second stage compares them against phonetic and numerical information learned from data-driven training. Fig.25 reiterates the analytical portion of Fig.16. It was crucial to consider both aspects to confirm if numerically derived patterns reach linguistic interpretability.

Hence, part of the goal was to check if a) and b) of Fig.25 remained consistent. This inquiry, however, was situated within the broader objective of understanding how the features encode non-categorical traits and whether discovered patterns exhibit linguistically relevant relationships. In this context, a) and b) were not entirely mutually exclusive; instead, their combined insights contributed to a deeper understanding of the properties of code vectors. In cases where interpretability from one perspective was unavailable or insufficient, we took reference from the other. Ultimately, the analysis is orchestrated toward defining the relationships of patterns below the categorical level. The task amounts to substantiation in Fig.28, as the first substitution error Z to S will be used for exemplary demonstration.

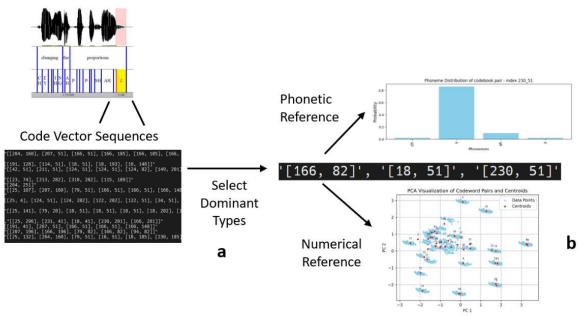


Figure 25. Sub-Segmental Pattern Analysis Schema

In **Z to S**, the three most dominant pattern indices representing its code vector sequences were: [166, 82], [18, 51], and [230, 51]. Pattern [18, 51] was an unavailable pair from the L1 TIMIT reference, and thus, we relied on individual codebook information, i.e., the 18th entry from codebook1 and the 51st entry from codebook2.

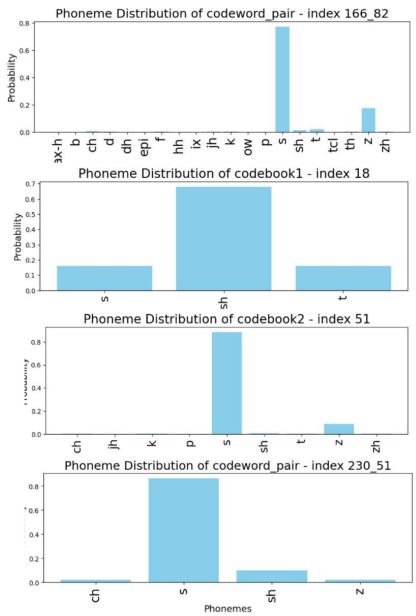


Figure 26. Phonetic Attributes of Z to S Patterns

The phoneme distribution of four codeword (pair) indices in Fig.26 reflects the general trend of the substitution. They are associated with (palato) alveolar fricatives, with the primary distribution aligning with the recognized phoneme S. The distinguishing feature lies in the secondary distribution: index [166, 82] corresponds to the targeted voiced counterpart Z, while [230, 51] aligns with the voiceless SH. Index [18, 51] occupies an intermediate position between these contrasting elements. Specifically, the 18th entry of codebook1 shows

the highest probability for SH, while the 51st entry of codebook2 exhibits similarities to [166, 82]. Notably, the phonetic interpretation of patterns reveals a gradient error continuum scaled by the assumed degree of changed articulatory value. The progression begins with the index [230, 51] that bears a less voicing characteristic and moves on to [166, 82], associated with the voiced Z. Pattern [18, 51] serves as a middle ground by being partially linked to both SH and Z. Its intermediary position leans more towards [230, 51] concerning their shared second codebook index and its strongest association with SH, which holds the secondary role in the distribution of [230, 51].

For the numerical comparison, we needed to borrow the first codebook vector space in addition to the joint space to plot the pattern indices. Although [18, 51] lacked an L1 counterpart for measuring distances as an index pair, it was comparable to [230, 51] based on its first codeword index. Accordingly, Fig.26 visualizes the Euclidean distance calculated in two vector spaces. For the paired indices, we utilized cluster IDs for visualization as 2022 individual data points were less tractable to directly position. The choice was for representational purposes as well as to assess the level of separability. The clustering result with the designated 39 categories sets the standard for comparison, paralleling the phonemic grid against which our non-categorical analysis is measured. Note that the precise calculation was based on the individual indices instead of cluster centroids.

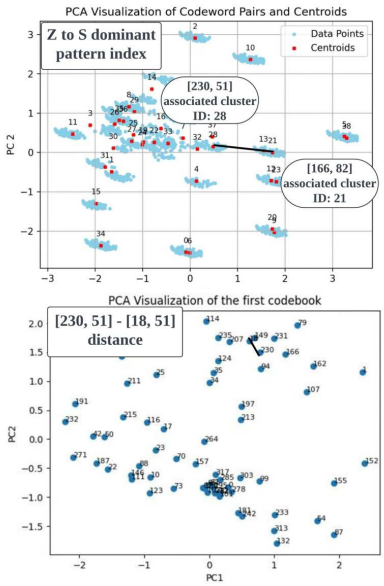


Figure 27. Eucledian Distance of Z to S Patterns

Finally, the information gathered from the two perspectives converges on the continuum scale shown in Fig.28. Phonetic interpretations are marked in green, while the numerical interpretation of pattern-wise Euclidean distances is shown in blue. The values were rounded to the nearest hundredth. Although the difference between [166, 82] and [18, 51] could not be calculated due to the absence of the latter codebook pairing, the general trend follows that the phonetically inferred positions align with the Euclidean distances.

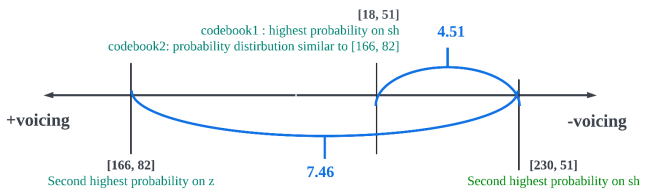


Figure 28. Discovered Patterns in Z to S substitution

Notably, three overarching characteristics define the pattern spectrum. First, as previously mentioned, 1) patterns exhibit a scaled relationship corresponding to the assumed degree of change in articulatory traits. The voicing identity driving the shift serves as the axis of the error continuum. Second, 2) the intermediary pattern is non-categorical. Index [18, 51] assumes ambivalent identities in two codebooks as the 18th and 51st codewords each bear values closer to [-voicing] and [+voicing]. The incoherent positioning shows that [18, 51] is associated with neither category, which is also presumed to be the cause of its absence in native speech. Finally, 3) patterns are skewed toward the closest approximates in L1. The tendency of [18, 51] to lean more toward the voiceless end, represented by [230, 51], reflects the difficulties arising from the L1 alveolar fricatives being restricted to voicelessness (4.2.1). These three characteristics were also prevalent across the following three substitution trajectories.

Error Type	DH to D	V to B	F to P
Pattern Index	[197, 155], [107, 120], [204, 120]	[197, 155], [197, 230], [204, 162]	[197, 155], [197, 284], [166, 196]

Figure 29. Pattern Indices of Fricative to Plosive substitutions

Fig.29 summarizes the discovered pattern indices of the second substitution trajectory that involves a shift from fricatives to plosives. Reflecting the continuous and non-continuous nature of two substitution categories, 1) patterns were scaled on the spectrum of continuity. At the [+continuity] end in Fig.30, the index [197, 155] of DH to D and V to B have high associations with fricatives. Meanwhile, at the [-continuity] end in Fig.31, [107, 120] of DH to D and [197, 230] of V to B are associated with plosives and silence markers. Progressions from continuous to non-continuous traits are mediated by 2) intermediaries in Fig.32 that assume [+continuity] value in the first codebook, while [-continuity] in the second codebook.

Note that the ranking is relative within each error continuum, as the same value may hold different rankings across error types. For example, the second index of the V to B intermediary [204, 162] shares distributions with approximants, which is non-continuous compared to the first index's primary association with vowels. Nevertheless, it may be more continuous compared to the second index of the DH to D intermediary [204, 120]. A common finding that cuts across relative rankings is that the larger the discrepancy, the more likely the pair is to appear less frequently in L1. Considering that value conflict reflects non-categoricality, we may interpret this causal relationship as stemming from the L2-specific nature of non-categorical pronunciation. [204, 120] displaying the largest gap in continuity value had only a single occurrence with nasal n, while the other two pairs showed greater presence. The association of the nasal here suggests its dual nature since the sound involves obstruction of airflow in the oral cavity yet unhindered in the nasal cavity.

We deferred the discussion on F to P to address the third characteristic of distributional asymmetry. Among three pattern indices of F to P, [197, 155], representing the [+continuity] end in the other two substitutions, exhibits the least characteristic of continuity. The other two indices, [197, 284](Fig.32) and [166, 196](Fig.30) are almost entirely associated with continuous fricatives. In other words, pattern distributions are more homogeneous towards the targeted value. This difference in dispersion rate translates to vector plotting in Fig.33. The triangulation of F to P patterns involves smaller between-pattern distances, forming a tight cluster around the most continuous [166, 196]. 3) The distributional asymmetry, where F to P patterns are considerably more skewed towards [+continuity] than DH to D and V to B, reflects the learners' difficulties in articulation. In 4.2.1, we explained that voiced fricatives pose a greater challenge than voiceless ones, as the Korean fricative inventory is limited to voiceless sounds,

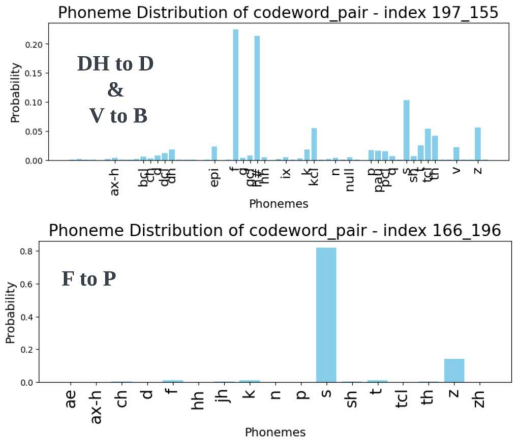


Figure 30. [+continuity] end in the Fricative to Plosive substitutions

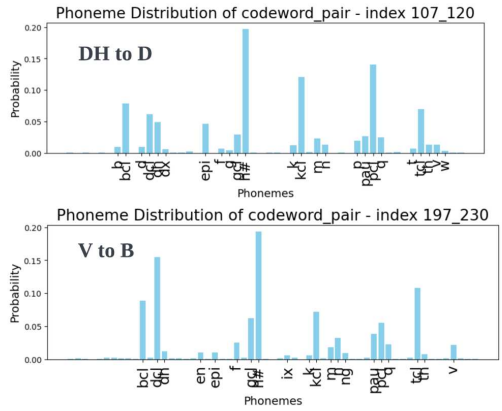


Figure 31. [-continuity] end in the Fricative to Plosive substitutions

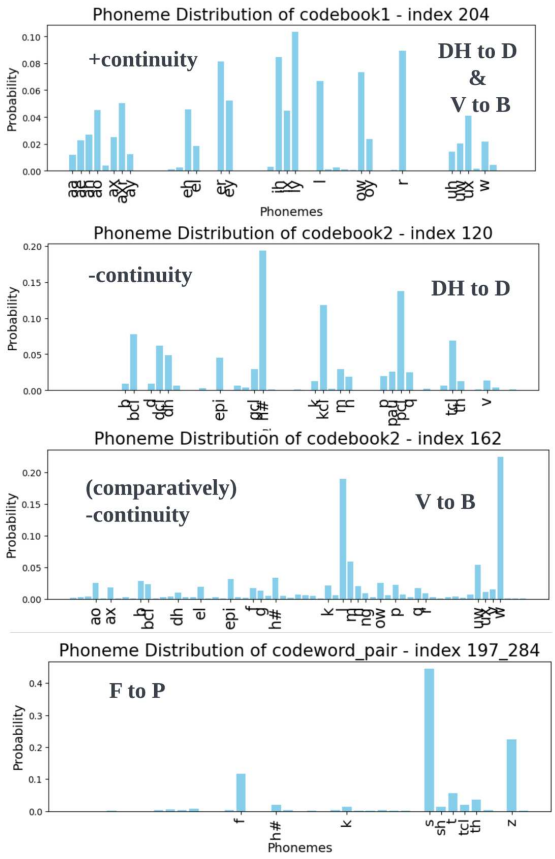


Figure 32. Intermediaries of the Fricative to Plosive error substitutions

and the overall sound system lacks a voicing distinction to provide articulatory familiarity. Hence, DH and V, which are more difficult to pronounce, incur greater dispersion as there is a higher likelihood for deviation. On the other hand, the relative ease of producing F brings its values close to the intended voicelessness. We may further understand the asymmetry in the context of cross-lingual influence when considering that pattern index [166, 196] and [197, 284] share most of their distribution with S, which is the closest approximate to the available fricative in Korean, i.e., /s/.

Fig.34 overlays the Euclidean distance data from Fig.33 onto the recurring dynamics moving from [-continuity] to [+continuity]. Because [204, 162] mediates between vowels and approximants, it is positioned closer to the [+continuity] end compared to [204, 120]. The smaller discrepancy leads to a higher frequency than [204, 120] of 502 occurrences noted in purple. Note that the visualization may not reflect precise values as it is based on cluster centroids and influenced by the loss of information from dimensionality reduction.

Fig.35 summarizes the discovered pattern indices of the third substitution trajectory concerning changes in the channel of airflow. The switch between lateral and rhotic liquids can be defined by differences in vocal tract configuration. The lateral liquid guides the airflow towards the side of the tongue by approaching the tongue tip near an alveolar ridge. This process of creating a lateral channel concerns the trait laterality and is absent in English alveolar approximant /ɹ/. The bunch r, in contrast, retracts and lowers the tongue tip, allowing air to flow over the center of the tongue. As the process involves a lesser degree of constriction and obstruction thereof, rhotic/ɹ/ are more resonant and acoustically closer to vowels. Thus, when the pattern shares more probability with vowels, we may interpret it as bearing

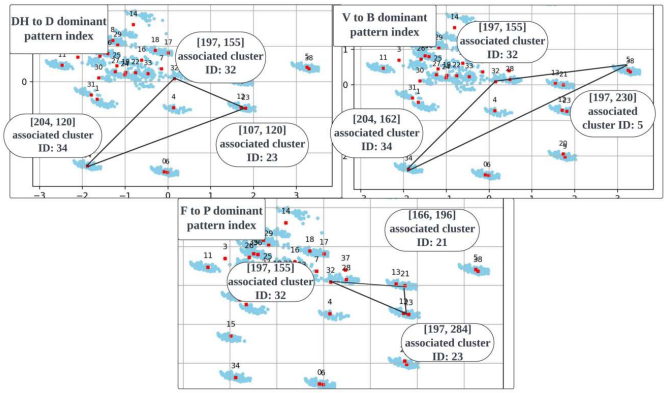


Figure 33. Eucledian Distance of Fricative to Plosive Patterns

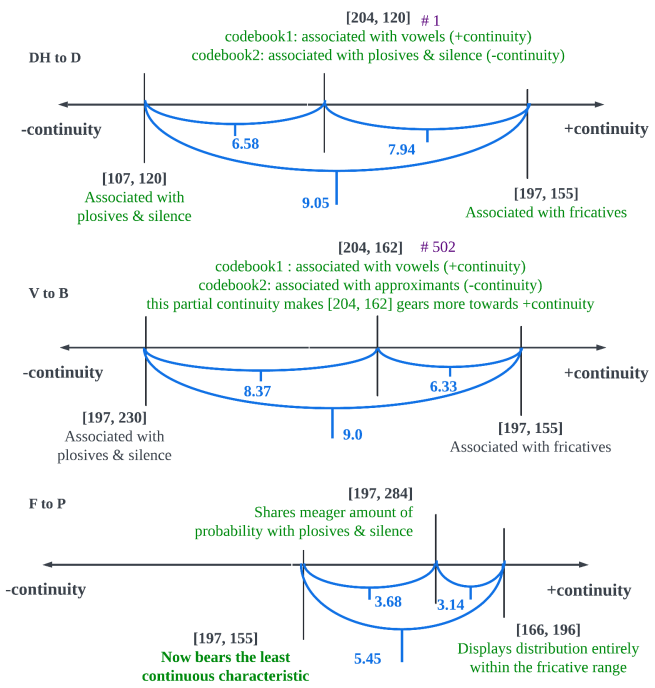


Figure 34. Discovered Patterns in Fricative to Plosive substitutions

a comparatively [-laterality] attribute. Given this understanding, 1) patterns were scaled on the laterality spectrum with native phoneme distribution moving from lateral association [+laterality] to rhotic and vowel association [-laterality].

Error Type	L to R	R to L
Pattern Index	[204, 162], [191, 162], [191, 212]	[22, 101], [191, 235], [204, 162]

Figure 35. Pattern Indices of Liquid substitutions

Fig.36 first shows the phonetic attribute of the [+laterality] end, which overlaps between two error directionalities. As we should see, pattern attributes share a common ground to a considerable degree. This is an expected result concerning they occur under the systematic interaction of the same L1-L2 pair. At the [-laterality] end in Fig.37,[22, 101] of R to L shares distribution mostly with rhotic and r-colored vowels. Although [191, 212] spots a less salient trait, it is more heavily associated with vowels than [191, 162].

Meanwhile, 2) the intermediaries [191, 162] and [191, 235] in Fig.38 display ambivalent characteristics by having comparatively higher [+laterality] values in the second to the first codebook entry. Reflecting the non-categorical nature, both paired indices record only 3 instances in TIMIT. The former was associated with L twice and UW once, while the latter was associated with IH, UH, and UW once each. Importantly, both belonged to cluster 15 in the numerical calculation of vectors. In Fig.39, the two pattern indices are mapped to the same cluster centroid. The second codebook vector space is additionally provided for detailed viewings of the data points. As [191, 162] and [191, 235] only differ by the second entry, their distance is plotted through the 191st and 235nd codeword alongside the 212nd codeword to demonstrate the overlap mentioned earlier. The adjacency of three indices reflects not only the shared non-categorical trait among intermediaries but that the [-laterality] end in L to R bears partial lateral characteristic as shown in Fig.37. That is, as [191, 212] gears

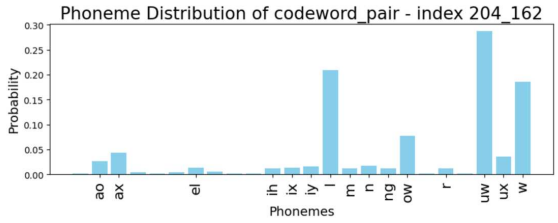


Figure 36. [+laterality] end in the Liquid substitutions

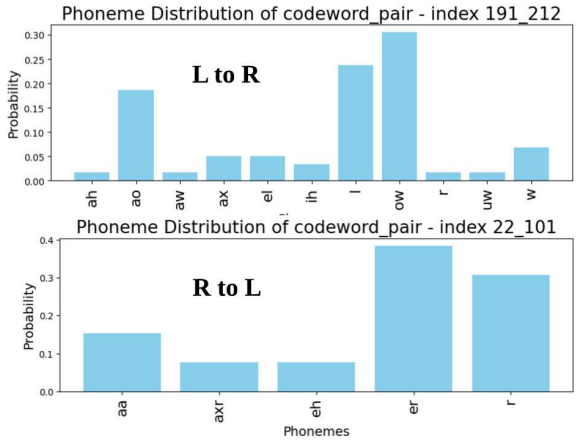


Figure 37. [-laterality] end in the Liquid substitutions

towards the intermediary, it forms a close acoustic distance with the lateral sound.

Hereafter, we spot the third overarching trait regarding distributional asymmetry related to the learners' L1. Along with the close distance [-laterality] end of L to R forms with the intermediaries, intermediaries themselves show proximity to the [+laterality] end. The laterality spectrum in the summary of Fig.40 coincides with the hierarchy formed along the vertical axis of the visualization and thus has been superimposed. 3) The tendency of both substitution patterns towards the lateral end reflects the relative articulatory difficulty between the two liquids. As explained in 4.2.1, the unobserved variety of rhotic /ɹ/ in the learners' L1 accompanies greater difficulties in production than /l/ present in both sound systems. The skewed distribution also reflects the funding in Koo (2012).

Thirdly, the three pattern indices constituting the **AH to AA** error were: [232, 131], [264, 131], [264, 6]. From a phonetic standpoint, the patterns are differentiated by the relative association ratio of AO compared to AA. Since AO shares the same vowel height as AA, 1) the pattern distribution is scaled by the change in vowel height. The gradual change is charted vertically in Fig.41. Among three indices, 2) The intermediary [264, 131] exhibits inconsistent positioning, with codebook 1 assigning a higher probability to AA than AO, while codebook 2 shows the opposite trend, assigning a higher probability to AO than AA. This non-categorical trait is presented in Fig.42.

Finally, we inspected the Euclidean distance among raw vectors and confirmed consistent understanding across two viewpoints. Fig.43 plots the pattern indices by their cluster IDs, where [264, 131] and [264, 131] are associated with the same cluster. Moreover, the 15th cluster that merges these two patterns is adjacent to the cluster ID of [232. 131]. Hence, the distance among the three patterns is not large, implying a shared attribute. This common ground corresponds to the gradient movement in Fig.42, where probability with AO is consistently present even at the [-high] end, i.e., [264, 6]. In other words, the shared phonetic distribution gears towards the [+high] attribute, which is consistent with the proximity of the merged cluster to the [+high] end pattern [232, 131]. Fig.44 summarizes the

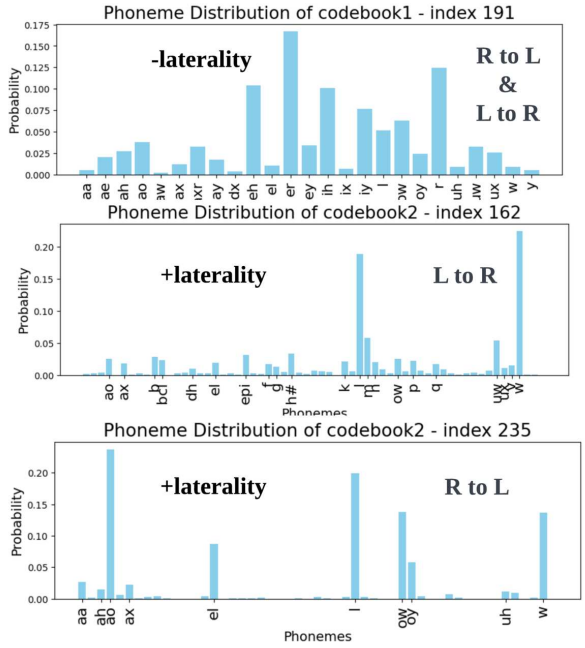


Figure 38. Intermediaries of the Liquid substitutions

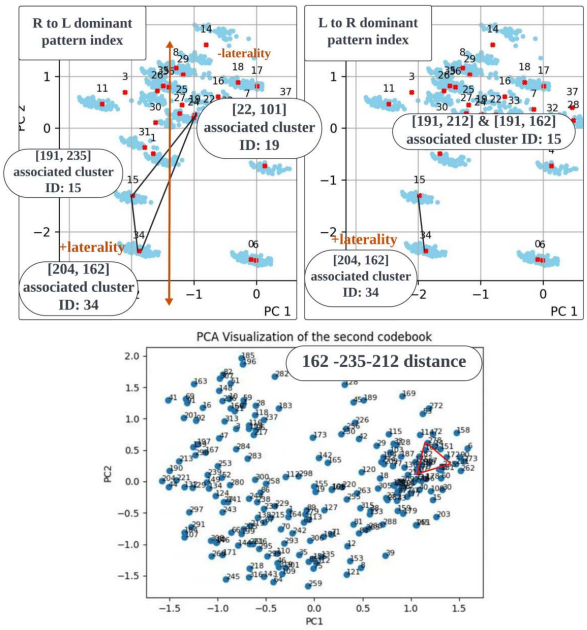


Figure 39. Euclidean Distance of Liquid substitution Patterns

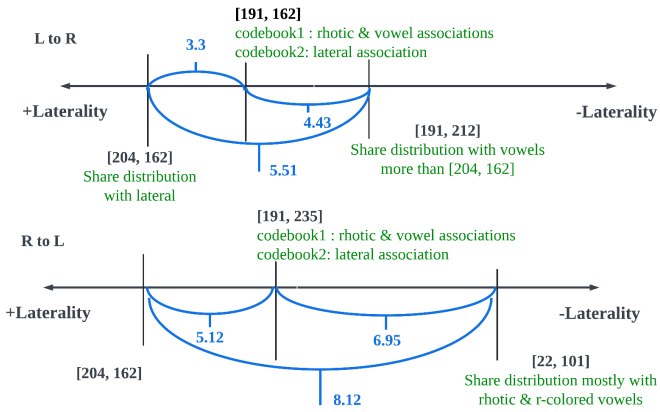


Figure 40. Discovered Patterns in Liquid substitutions

converged understanding with precise calculated values marked on the error continuum. 3) the asymmetrical distribution skewed to the [+high] end of the substitution spectrum aligns with the finding of Koo & Oh (2001) that Korean learners tend to pronounce AA near the mid-vowel region. Accordingly, we have mentioned in 4.2.1 that AH to AA patterns will likely lean towards the AH category.

The final two substitution trajectories required a different analytical approach because most of their pattern indices lacked corresponding examples in L1. Namely, they assumed conflicting pairings throughout all instances, a characteristic previously limited to intermediary patterns. As analyzed earlier, the discrepancy in two codebook values leads to rarity in occurrences. To address the absence of paired references thereof, we examined individual codewords, similar to the approach used for intermediary analyses in the first four trajectories. Fig.45 summarizes the discovered pattern indices of the tenseness substitutions, whose codeword values are calculated in Fig.46. The paired indices utilize three entries from codebook1 and two entries from codebook2. The codeword set from each codebook converges on the common characteristic of bearing tense and lax values, respectively. The EH to AE patterns are excluded from the individual analysis as they were the exception to the rarity rule. We will elaborate on their unique status in the discussion chapter.

Tenseness values were derived by calculating the probabilistic ratio of associated tense phonemes to their lax counterparts. The second row of the table in Fig.46 lists the tense and lax phonemes used in the calculation. The sum of the co-occurrence probability with tense phonemes was divided by the total co-occurrence probability of lax phonemes. A graphical example illustrating the calculation for the 42nd codebook entry is shown below. Here, the phonemes boxed in red lines represent the lax set, whereas the blue lines represent the tense set. The fourth row of the table summarizes the tense-to-lax ratio of two codebooks, where one can notice that the first codeword entry overall exhibits a lower value than the second. The phonetic understanding of this metric is reflected in the vector-wise Euclidean distance. In Fig.47, we scaled the overall phonetic attributes of the paired index by aggregating the ranking in two codebooks. Pattern positioned by the relative size of the tense-to-lax ratio of two codewords aligned with numerical interpretation in blue font.

We omitted visual plotting in the joint space since the instances were extremely rare to derive a comprehensive outlook. Namely, [22, 268] showed three occurrences (once with EH and twice with EY), [22, 234] was not present in TIMIT, [191, 234] recorded four instances (twice with IY and twice with EY), and [42, 234] has single co-occurrence with EY. Nevertheless, cluster memberships of L1-present indices were recorded for the analysis of EH to AE.

The EH to AE substitution utilizes codebook pairings well-represented in native speech, thus departing from the other three substitutions. Its pattern indices [264, 135], [264, 64], and [264, 315]

belong to the 8th and 29th clusters, which are adjacent to each other. Hereafter, the plotted locations in the vicinity exhibit a coherent trend. When compared against the three cluster centroids of the tenseness spectrum in AE to EH (Fig.47), the distance is the shortest with the most lax 24th cluster, while the longest with the most tense 15th cluster. Moreover, the intermediary 11th cluster also takes a middling position in length. The bottom graph of Fig.48 delineates these between-cluster distances that increase proportionately to tenseness. Consequently, we may expect the patterns to gear toward the [-tense] end of the spectrum. Such tendency is reflected in the phonetic attributes in Fig.49 that vertically charts the gradual change from the [-tense] end [264, 6] to the [+tense] end [264, 315]. [264, 6] is initially associated solely with lax EH, but as we move down the index graph, its proportion gradually accommodates tense AE, along with other tense phonemes defined in Fig.46. Even the most tense pattern [264, 315], however, shares a considerable portion of its probability with lax vowels including EH. The situation is comparable to the skewed distribution in AH to AA, where the [-high] end pattern shared probability with the representative phoneme of the [+high] end, AO. Resultingly, we arrive at the distributional asymmetry marked in the top error continuum of Fig.48.

Bringing the four error continua of Fig.47 and 48 together, the difference in dispersion rate between two substitution directionalities follows the purported articulatory difficulty in 4.2.1. Errors incurred by the more challenging production target have their patterns more dispersed, bearing values distant from the intended trait. In contrast, errors of the opposite directionality with the easier target show a more condensed distribution toward the targeted value. We previously observed this dispersion rate disparity in the fricative to plosive examples. The same asymmetrical dynamic applies to the substitution between IH-IY and EH-AE pairs. In IH to IY and IY to IH, lax to tense error incurs more dispersion, while tense to lax patterns are more aligned with the canonical sound. In AE to EH and EH to AE, tense to lax error incurs more dispersion, while lax to tense patterns are more aligned with the canonical sound. This distribution trend relates to lax IH and tense AE being the more difficult segment among the tense-lax pair (Yang, 2010; Tsukada et al., 2005).

The final substitution pattern of diphthong reduction is attributed to the failure to execute rapid articulatory movements. As explained in 4.2.1, diphthongs involve movement between two articulatory positions. When this movement is inadequately executed, the production becomes segmented, often resulting in substitution by adjacent monophthongs. Therefore, a metric that measures tongue movement is relevant for examining the substitution pattern. Previous studies, including Oh (2013), have used the F2 formant value for this purpose. Since F2 reflects the front-back positioning of the articulator, we can measure the movement distance by observing shifts in its value. A larger shift indicates more drastic movement, which is associated with segmentation and an increased likelihood of substitution. In this context, successful articulation of the diphthong depends on the coherent positioning of the tongue along the front-back axis.

Accordingly, we calculated the resolved front/back degree of two codebook phoneme distributions as an indicator of monophthongization possibility. Since the diphthong nucleus is the focal point of production, the direction of resolution depends on its position within the articulatory movement. Both OW and EY involve an inward shift from a nucleus positioned relatively outward. As a result, the front vowel EY undergoes front-to-back movement, while the back vowel OW undergoes back-to-front movement. Therefore, **OW to AO** patterns were analyzed based on the resolved degree of backness, represented by the back-to-front ratio in Fig.50. In contrast, **EY to EH** patterns were analyzed based on the frontal resolution standard, quantified by the front-to-back ratio metric.

The values for both metrics were derived using the tense-to-lax ratio calculation scheme explained earlier. Referring to the front and back phonemes listed in the table, the sum of the co-occurrence probability of back phonemes was divided by that of front phonemes to compute the back-to-front ratio. For the front-to-back ratio, the order was reversed, comparing the probabilistic ratio of associated front phonemes to their back phoneme counterparts. Instead of limiting positional identification to vowels, all sounds involving the front and back cavities were considered.

The calculated values in the fourth and sixth rows demonstrate a non-categorical nature, a cause of the lack of L1 representatives. Dominant pattern indices representing substitution sequences were [191, 212], [232, 212], and [22, 212] for OW to AO and [22, 234], [42, 234], and [191, 234] for EY to EH. Constitutive codewords of these pairings were the target of the metrics calculation. As can be noted, codewords from the first codebook generally display traits of front vowels, while those from the second codebook exhibit values closer to back vowels. Such conflicts in articulatory positioning likely led to phonetic variations not present in L1. Because categorical judgment overlooks these subtleties, phonemic recognition broadly classified them as monophthongs. We can further examine how code vectors encode these variations by analyzing how the ranking of positional conflicts translates into their numerical identities.

Fig.51 compares the relationship between OW to AO patterns from two standpoints: 1) scaled by the phonetic understanding of positional conflict, and 2) defined by the Euclidean distance in the joint space. The top error continuum uses the back-to-front ratios of individual codewords in Fig.50 to rank pairings according to their resolved degrees of backness. Since frequencies and the level of contradiction are inversely proportional, the most coherent pairing, [191, 212], records the highest occurrences, where the gap between the two codebook values (1.687 and 2.084) is much smaller than the gaps for the other two indices. [232, 212] and [22, 212] are phonetically similar, as they resolve the first codebook value of 0.7 and 0.599 with the second codebook value of 2.084. Consequently, their frequencies are nearly identical, with [232, 212] slightly surpassing [22, 212].

This frequency difference aligns with the vector plotting at the bottom, where the 27th cluster (associated with [232, 212]) is sufficiently apart from the 15th and 24th clusters (associated with [191, 212] and [22, 212], respectively). The latter two vectors are adjacent, reflecting the near-equivalent level of positional conflict. This finding suggests that code vectors are numerically aware of the phonetic identity associated with the articulatory traits of diphthongs.

Moreover, there is a correlation between diphthongal properties and tenseness. The cluster previously defined to constitute the tense end of the spectrum (15) in Fig.47 is associated with the most resolved pair exhibiting the closest value to a diphthong. Conversely, the cluster positioned at the lax end of the spectrum (24) is associated with the least resolved pair bearing the closest value to monophthong. This consistency highlights the versatility of the code vector understanding and is further supported by the analysis of EY to EH.

Fig.52 compares the phonetically derived relationships of EY to EH patterns with the tenseness substitution patterns examined earlier. In the topmost continuum, patterns are scaled by aggregating the frontness ranking in two codebooks. Pattern [42, 234] forms the resolved end of the spectrum with the highest front-to-back ratio pairing (2.666 - 11.363). This same index also serves as the intermediary pattern in the IY to IH spectrum displayed below. Meanwhile, the intermediary of EY to EH [22, 234] bears less characteristic of tenseness than [42, 234] in IH to IY. Lastly, the pattern least resolved towards frontness [22, 319] belongs to the 24th cluster, which is associated with the most lax patterns in IH to IY. Referencing the yellow connection line, one can observe that the ordering of the tenseness transfers to the frontal resolution degree. The more the pattern aligns with the value of diphthongs, with its production focus centered around the nucleus, the closer it is to the tense end of the continuum. This comparison was possible as patterns of front vowel substitutions overlapped considerably, a discussion topic for the following chapter. Thus, similar to tenseness analysis, we spared the visualization in vector space as

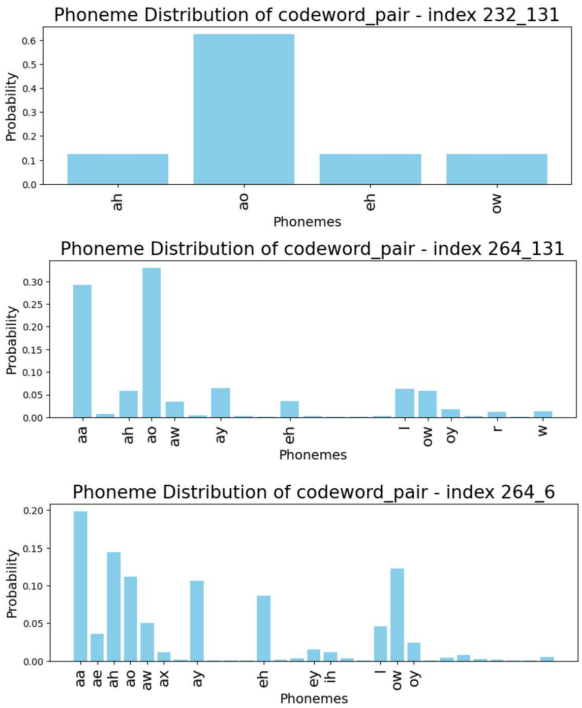


Figure 41. Phonetic Attributes of AH to AA Patterns

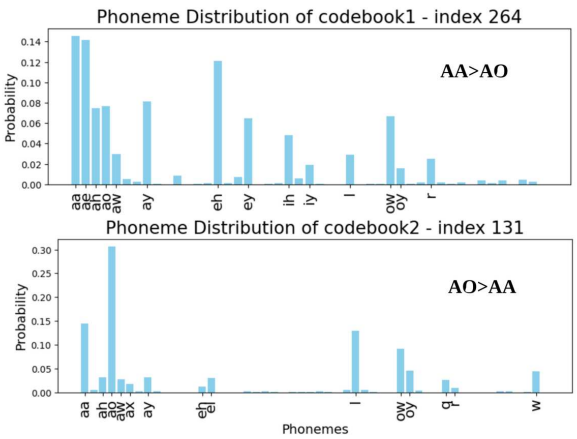


Figure 42. Intermediate Pattern of AH to AA

the pattern indices lacked representative samples.

5. Discussion

In 4.1, we confirmed that L2 speakers use code vectors differently from L1 speakers. This difference arises not only from variations in feature usage frequency, as demonstrated by the frequency vector clustering (Fig. 17-8), but also from the distinct inventories employed by the two groups. Notably, the range of featural diversity expanded with language proficiency, with the amount used by native speakers setting the upper limit (Fig. 23). As greater overlap with L1 speakers corresponds to increased articulatory adeptness (Fig. 22), we expect that the more pronunciation deviation there is, the higher the chance for the L2 speech to utilize varieties not present in L1.

In 4.2.2, discrepancies between paired codeword values led to L2-specific non-categorical traits. The non-categorical patterns served as intermediaries in fricative, liquid, and vowel height substitutions. In tense-lax and diphthong reduction errors, the trait spanned the entire pattern spectrum. These conflicting pairings over the changed

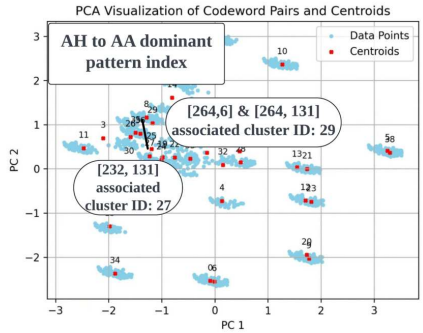


Figure 43. Euclidean Distance of AH to AA Patterns

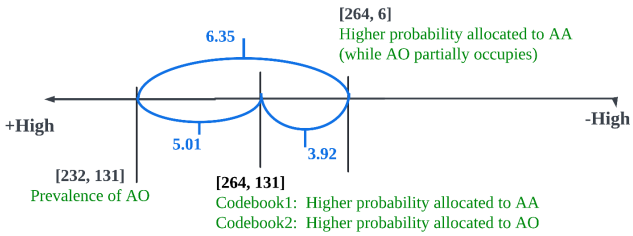


Figure 44. Discovered Patterns in AH to AA substitution

Error Type	IH to IY	IY to IH	AE to EH	EH to AE
Pattern Index	[22, 268], [22, 234], [191, 234]	[22, 234], [42, 234], [191, 234]	[22, 268], [42, 234], [191, 234]	[264, 135], [264, 64], [264, 315]

Figure 45. Pattern Indices of Tenseness substitutions

Vowel Type	Tense	Lax
	ae, aw, ay, ey, iy, ow, oy, uw	aa, ah, ao, ax, axr, eh, er, ih, ix, uh, ux
Tense-to-Lax ratio	Codebook1	Codebook2
	191 (0.511) > 42 (0.375) > 22 (0.315) [-tense]	234 (9.744) > 268 (2.981) [+tense]

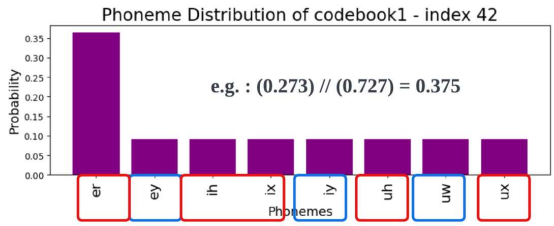


Figure 46. Calculation of Tenseness Value

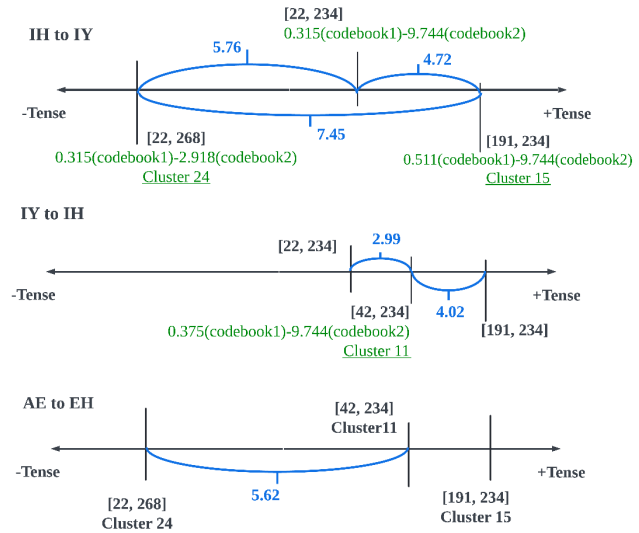


Figure 47. Discovered Patterns of Tenseness substitution (excluding EH to AE)

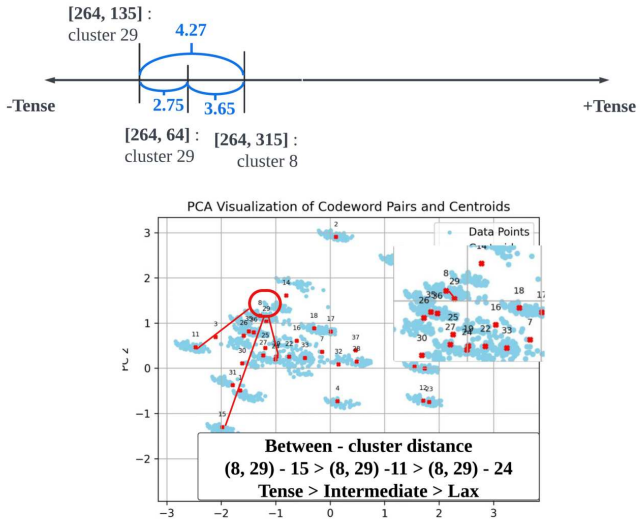


Figure 48. Eucledian Distance of EH to AE Patterns

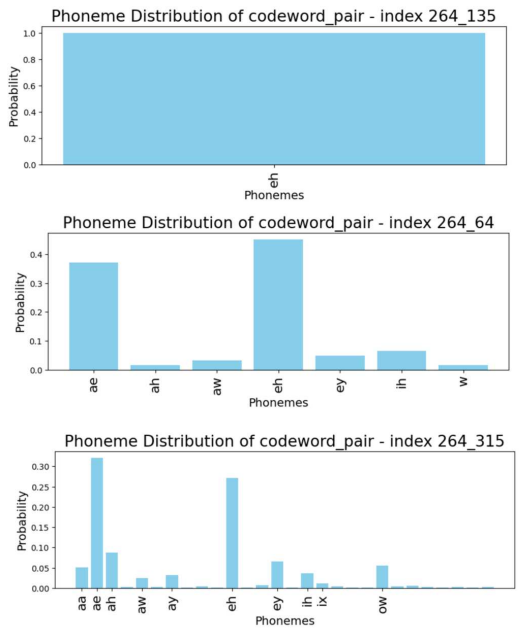


Figure 49. Phonetic Attributes of EH to AE Patterns

Phoneme Type	Front	Back
	eh, er, ey, ih, iy, y, l, ix, el, dx, b, bcl, f, p, n, m, v, dcl, dh, pcl, tcl, t, z, nx, em, th, s	aa, ae, ah, ao, aw, ow, uh, uw, ay, oy, w, ux, r, hh, q, k, hv, ng, g, gcl
Back-to-Back ratio	Codebook1	Codebook2
	191 (1.687) > 232 (0.7) > 22 (0.559) [-back]	212 (2.084) [+back]
Front-to-Back ratio	Codebook1	Codebook2
	42 (2.666) > 22 (1.788) [-back]	234 (11.363) > 319 (3.606) [+back]

Figure 50. Calculation of Front/Backness Resolution

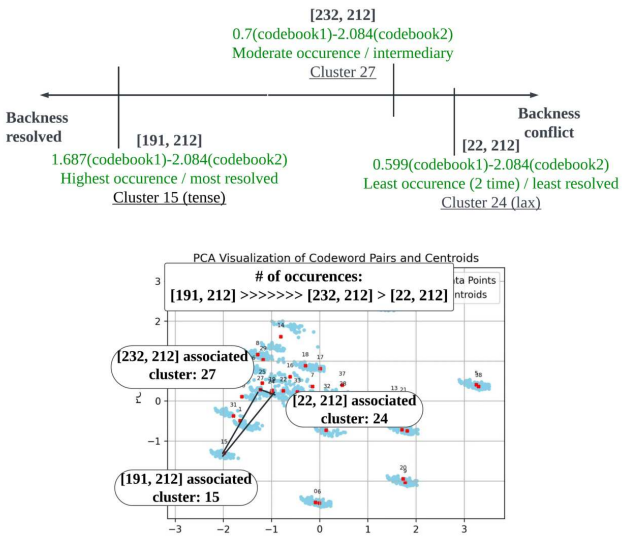


Figure 51. Pattern Distribution of OW to AO

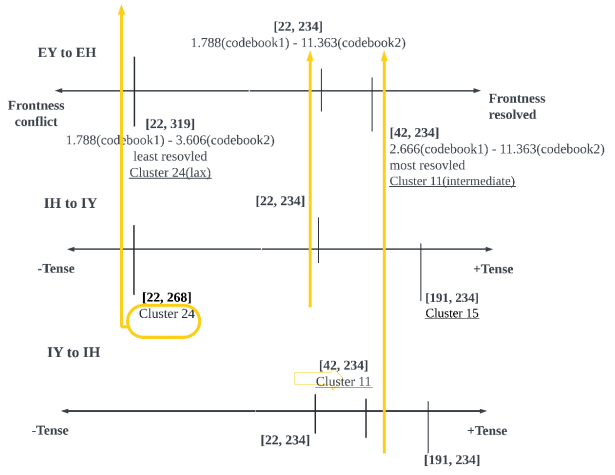


Figure 52. Discovered Patterns in EY to EH substitution

articulatory trait instantiate non-overlapping L2 code vector inventories compared to L1, outlined in 4.1. Thus, we question whether there are underlying factors rendering the latter two trajectories particularly vulnerable to pronunciation deviation. Intuitively, vowels exhibit greater variability than consonants, which may make their error patterns more likely to involve L2-specific inventories than the substitution of Z, DH, V, F, L, and R. Beyond this general conjecture, a common denominator of errors involving pattern indices rarely present in L1 was that they concerned front vowels.

At the end of 4.2.2, we briefly mentioned that tense-lax substitution patterns overlap with those involved in front diphthong reduction. Notably, our analyzed L2 front vowel patterns converged on two overarching characteristics: overlap and rarity. The left table of Fig.54 reiterates their indices, with L1 frequency annotated in parentheses. The two traits suggest that front vowel errors form a cluster distinct from L1 renditions in the vowel space. This interpretation aligns with the articulatory analysis in Koo & Oh (2001), which reports that the Korean English vowel space is narrower than that of native American speakers, particularly in the frontal region. Accordingly, Korean speakers are not versed in manipulating the oral constriction required for the accurate articulation of English front vowels. Tongue tip movement is more limited, creating an acoustic distance between the IH, IY, EH, EY, and AE of learners and native speakers.

This peculiarity of front vowel dynamics was not only present within the L2 data but was also reflected across the code vector distributions of L1 and L2. To analyze the acoustic distance outlined above, we calculated the Euclidean distance between the indices most representative of front vowel segments in L2 erroneous speech and those in L1 TIMIT. The pattern indices of vowels in TIMIT were derived using the same method explained in 3.2. The representativeness was determined based on frequency, as the second column of the left table highlights selected indices from our surveyed L2 patterns. The third column annotates the most frequently occurring pattern index from L1, alongside the fourth column, which calculates the distance. To compare the relative size of this measure, we further calculated the L1-L2 code vector distance for the other two back vowels we analyzed. The right table mirrors the scheme of the left table, with the calculation target being the vowels AH and OW.

Overall, the distances between the representative index of the L1 phoneme and the corresponding erroneous segment in L2 are smaller in back vowel substitutions (5.62, 5.26) than in front vowel pairs (8.26, 7.65, 4.95, 7.98, 8.53). This difference reflects the greater deviation occurring in the front vowel space. Back vowel patterns were also more frequently observed in native speech. Although OW to AO exhibit conflicting identities, [191, 212] had a corresponding sample size of 59. The correlation between frequency and Euclidean distance from the L1 dominant index suggests that acoustic proximity contributes to better correspondence. Moreover, back vowel indices do not experience the overlap present in front vowels. The effect on vowel space limits is, therefore, coherent: a restriction in frontal cavity movement leads to the use of L2-specific features due to discrepancies with L1 acoustics. This, in turn, blurs sound contrasts and hinders distinctive production across native phoneme categories, a phenomenon confirmed by the code vector overlap.

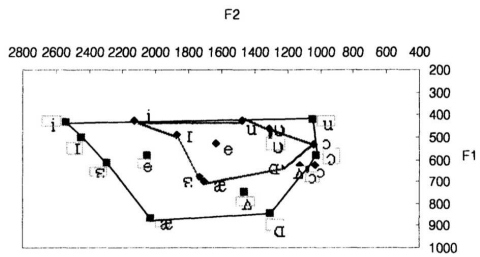


Figure 53. Vowel Space Analysis in Koo & Oh (2001). plotted formant values were averaged across speakers, represented by the diamond (Koreans) and square (Americans) symbols

	L2 error segment	L1 canonical segment	L1-L2 Euclidean distance
IH	[22, 234] (unobserved), **[191, 234]** (4), [22, 268] (1) (from IH to IY)	[207, 81]	8.26
IY	[191, 234] (4), [22, 234] (unobserved), **[42, 234]** (1) (from IY to IH)	[207, 205]	7.65
EH	**[264, 64]** , [264, 135], [264, 315] (from EH to AE)	[207, 143]	4.95
EY	[42, 234] (1), [22, 234] (unobserved), **[22, 319]** (3) (from EY to EH)	[207, 268]	7.98
AE	**[22, 268]** (1), [42, 234] (1), [191, 234] (4) (from AE to EH)	[152, 8]	8.53

	L2 error segment	L1 canonical segment	L1-L2 Euclidean distance
AH	[232, 131] , [264, 131], **[264, 6]** (from AH to AA)	[207, 6]	5.62
OW	[22, 212] , [232, 212], **[191, 212]** (from OW to AO)	[207, 212]	5.26

Figure 54. L1-L2 code vector Euclidean distance of Vowel Errors

The EH to AE patterns, recording the shortest distance (4.95) from the L1 representative, were an exception to this trend. Their unique status was previously noted in 4.2.2, where the pattern indices had sufficient L1 correspondents, unlike other tenseness substitutions. Two defining traits contribute to this status: 1) immunity to both overlap and rarity and 2) the cluster merger (Fig.48). In the graphical illustration of Fig.48, clusters 8 and 29, associated with these patterns, were markedly adjacent. Since clustering into 39 categories mirrors the concept of the phoneme grid, this tendency to merge implies that sound segments are being recognized under the same category. That is, code vectors encode variations of EH to AE more leniently than the sporadic distribution of AE to EH (Fig.47). This observation is comparable to the findings of Yang (2013).

When Yang (2013) measured the relative distance between Korean English and American English front vowels, the production of AE and EH exhibited opposite trends across the two speaker groups. Referring to the vowel plot in Fig.55, one can observe that pronunciation variation in AE is greater for American speakers (blue) than for Korean speakers (emboldened black), whereas the reverse holds for EH. This difference in comparative dynamics leads to two potential outcomes. Given that the model is pre-trained on American English LibriVox data, the encoding of code vectors will reflect the production distribution of American AE and EH. Thus, 1) the greater L1 variability of AE will broaden its receptive field, creating more room for L2 error pattern correspondence. This explains why the

code vectors used in EH to AE had sufficient native speech presence. Furthermore, 2) the larger standard deviation in the training data makes the code vectors more lenient toward differences in sounds recognized as AE. As a result, their numerical encodings converged patterns to adjacent clusters.

Conversely, the receptive field of EH is narrower than the pronunciation range of Korean learners. Because the data distribution of EH is more consistent, code vector encodings trained on this information will have less room for pattern correspondence and impose a stricter definition of the acoustic attribute of EH. Consequently, AE to EH patterns had lower native speech correspondence while being more dispersed in the scheme of native phoneme categorization.

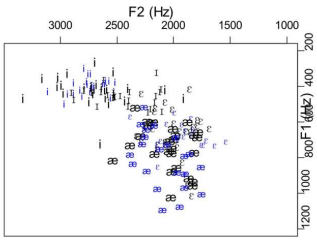


Figure 55. F1 and F2 of front vowels produced by Korean and American speakers in Yang (2013). American speakers are plotted in blue, whereas Korean speakers are in emboldened black.

Thus far, we have extensively discussed how the code vectors encode non-categorical traits by probing error samples that fully embody categorical conflict. The central thread of our analysis was the extent to which acoustic information statistically learned from unlabeled training overlaps with linguistic interpretability. We first found that deviations escaping categorical boundaries are exemplified by inconsistent pairings of codeword values and that these index pairs were rare in L1. Importantly, errors prone to adopting the L2-specific inventory involved articulatory targets confirmed to be difficult for Korean learners. On top of how these noteworthy observations relate to linguistic understanding, we will conclude this chapter by summarizing the linguistic relevance of the general findings.

In 4.2.2, we highlighted three overarching characteristics defining the substitution spectrum of fricative, liquid, and vowel height errors: 1) the gradient nature of the articulatory trait being altered, 2) intermediary patterns that are partially associated with both endpoint categories, and 3) distributional asymmetry that concurs with experimental results from existing L2 literature. Under this expository benefit, between-categorical positions of errors could be numerically calculated, as pattern vector distances reflected their phonetic status.

The substitution of tenseness and diphthongal properties formed a special case, where patterns were entirely composed of non-categorical L2 inventories. Hence, pair-level analysis was not possible, as we alternatively resorted to entry-level analysis. After calculating the value of substituted articulatory traits in individual codewords, patterns were ranked by aggregating two values. The inter-pattern relationships derived here reflected not only the numerical distance but also a comprehensive understanding of different phonetic traits. In OW to AO, code vector frequency concurred with cluster plotting (Fig.48), demonstrating that the feature is numerically aware of the cause of non-categoricality. In EY to EH, tenseness ranking correlated with the phonetic difference between the diphthong and its corresponding monophthong (Fig.52). The fact that phonetic understanding can be inferred even from entry-level viewing speaks to the diverse range of information code vectors can capture.

6. Conclusion

This work recreated the unsupervised L2 error pattern discovery experiments of previous literature using the self-supervised learning (SSL) representation, the Wav2Vec2.0 code vector. Leveraging its non-prescriptive and granular nature, the goal was to parse the phonetic variations of L2 errors that transcend the categorical definition.

The primary motivation behind our experimental design was to preserve the conceptual integrity of unsupervised discovery, thereby enabling the description of non-categorical traits free from top-down stipulations. This involved two key tasks: 1) confirming that the feature encodes L2 acoustics differently from L1, and 2) using its frame-wise encoding sequences to extract representative patterns of segments sharing the same categorical definition.

From the code vector inventory probing, we verified L2 discernibility of the features, which emerges from both the frequency of their usage and inventories uniquely present in L2 speech. In pattern discovery, we further found that the observed encoding unit differences result from an unlikely combination of codebook entries in L2. Namely, the two codeword vectors assumed conflicting identities at the opposite ends of the error pattern spectrum; If one displayed a positive value of the changed articulatory trait, the other displayed a corresponding negative value. This conflict instantiated L2-specific non-categorical traits, with increased usage of the relevant index pairs coinciding with a higher degree of phonetic divergence.

Crucially, patterns pertaining to a segmentally identical error displayed a gradient relationship along the changed articulatory value. In Z to S, the second highest probability in native phoneme distribution moved from sh[-voicing] to z[+voicing]. In DH to D, V to B, and F to P, native phoneme distribution moved from plosive and silence[-continuity] to fricatives, vowels, and approximants[+continuity]. In L to R and R to L, distribution moved from having more lateral[+laterality] to rhotic[-laterality] association (or vice versa). In AH to AA, the association ratio moved from allotting a higher probability to AA[-high] to AO[+high]. The above four substitution trajectories had sufficient correspondents in L1, as non-categorical pattern indices mediated the two ends of the gradience.

While direct co-occurrence-based analysis was infeasible for tense-lax and diphthong errors due to insufficient L1 sample sizes, analyses by individual codewords revealed a similar trend. Patterns could be scaled based on how two entries jointly render the desired target value. In tenseness substitution, probabilistic ratio of tense-to-lax phonemes formed the axis of the error continuum. Pattern indices typically bore the value of [-tense] in codebook1 and [+tense] in codebook2, which proved that it was the non-categorical trait that caused their rarity. In diphthong reduction, articulatory focus on the diphthong nucleus formed the distributional axis. The less resolved two entries were with respect to front/backness, the higher the chance of substitution, as this deviation also led to infrequency.

If correlation between pattern relationships and the articulatory traits responsible for changes outlines the linguistic relevance of code vectors, their distributional asymmetry further strengthens it. The resulting sub-segmental pattern distribution mirrored the production difficulties described in experimental studies on second language learning, with patterns skewed toward the sound closest to the available variant in the learners' L1. This asymmetry was observed in two ways: 1) the dispersion rate across different substitution types, and 2) within-substitution variation.

For the former, F to P involved smaller between-pattern distances than voiced fricative to plosive errors, with distribution concentrated towards the intended [+continuity] end. A similar trend was observed in the bidirectional dynamics of tense-lax substitution, where the more familiar targets IY and EH had their error patterns less dispersed. The skewed distribution within substitution could be found in the case of voicing identity substitution (Fig.28), liquid substitution (Fig.40), and vowel height shift (Fig.44), which each leaned towards the [-voicing], [+laterality], and [+high] end.

Importantly, these cross-lingual influences were numerically reflected in the featural values. Patterns were initially analyzed from a phonetic standpoint by examining co-occurring phonemes in L1. The derived pattern distributions were then examined numerically by cal-

culating the Euclidean distance, as each pattern was represented by a dominant code vector index. The two understandings overlapped, centering around phenomena narrated in previous L2 studies. This highlights the utility of code vectors in quantifying variations without requiring extensive linguistic expertise while also raising questions about their computational limits.

After all, code vectors were conceptually well-suited for unsupervised discovery, as they encapsulate language properties based purely on distributional patterns. By extensively exploring the range of variations they can capture, this study contributes to the practical domain of computer-assisted pronunciation training (CAPT), suggesting an alternative unsupervised feature to address challenges in sub-segmental labeling. On a theoretical level, their quantifiable phonetic relevance invites reflection on the nature of language learning and the extent to which established linguistic knowledge can be accounted for by non-human methods.

Notes

This paper is an adaptation of a manuscript accepted for publication in Phonetics and Speech Sciences (Vol.16, No.4). It also builds on the author's 2024 master's thesis, incorporating further developments after the thesis defense. The paper is primarily written by the applicant with minimal external involvement. It was selected for submission as it most comprehensively represents the author's past efforts in relation to the prospective research interest.

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